

# XRB-pilot workflow report

Data acquisition, processing, data-analysis methods and machine learning  
for black hole versus neutron star classification with RXTE, NICER and MAXI

October 2026 · describes repository `Chipwwwwww/XRB-pilot` at commit `666c2e0`  
(analysis rounds v1–v13)

**What this document is.** A step-by-step description of how the project works.

- which archives and catalogues are queried, and how files are downloaded and logged;
- how observations are selected;
- how spectra, light curves and event files are turned into features;
- which machine-learning models are trained, and how;
- how they are evaluated, and how everything is re-run.

Every rule, parameter and formula was read from the code (`scripts/`, `config.py`). Where useful, it was cross-checked with the technical report `report_en/main.pdf`. The results themselves are covered in the paper draft and the summary report; they appear here only as sample sizes.

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# 1 Overview of the workflow

## 1.1 The task

Each X-ray binary in the project is labelled from external evidence:

- *black hole* (BH): a dynamical mass measurement;
- *neutron star* (NS): type-I X-ray bursts or pulsations.

The workflow turns single X-ray observations of these sources into feature vectors, using:

- RXTE/PCA spectra and light curves;
- NICER/XTI event files;
- MAXI/GSC daily light curves.

Classifiers are trained on the feature vectors. Their scores are always computed for sources that the model did not see in training, and they are evaluated with statistics that treat the *source*, not the observation, as the independent unit. Everything is driven by Python scripts. HEASoft is used only where no Python equivalent exists: the XSPEC cross-check, the HI4PI  $N_{\text{H}}$  look-up and the NICER 3C50 background.

## 1.2 The workflow at a glance

Figure 1 shows the whole chain, and Table 1 lists the stages with their scripts and outputs. The chain has five parts, each described in later sections:

1. *External databases* (Section 2): source catalogues, observation catalogues and data archives.
2. *Sample construction* (Section 3): which sources, and which of their observations.
3. *Processing and feature extraction* (Sections 4 and 5.1).
4. *Data model* (Section 6): stored tables and the matrices given to the models.
5. *Modelling and evaluation* (Sections 7 and 8).

Section 9 covers running, logging and reproduction.

**Table 1:** Workflow stages, the scripts that implement them and their main outputs. Script numbers are those in `scripts/`; RXTE and NICER have parallel chains.

| Stage                          | What happens                                                            | Scripts                        | Main outputs                                                          |
|--------------------------------|-------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------|
| Archive inspection             | check the RXTE archive layout and file formats on two spectra           | 01, 05                         | <code>reports/step2_fits_check.md</code>                              |
| Source tables                  | label lists to positions; match to observation catalogues               | 03, 13, 20, 28, 35, 38, 41, 47 | <code>data/*/sources.csv</code>                                       |
| Candidate ranking              | eligible observations, time-stratified groups, seeded order             | 03, 23, 41, 47, 55, 60         | <code>data/*/observation_candidates.csv</code>                        |
| Retrieval + processing (RXTE)  | Standard Products, Standard-1, HEXTE, event mode; quality rules         | 04, 21, 24, 19, 26, 29, 42, 43 | <code>data/*/processed/features.npz</code> , timing tables            |
| Retrieval + processing (NICER) | TAP, prefix/tail/streamed event files, features, 3C50                   | 48, 52, 56, 58, 61, 62         | <code>data/v9-v13/processed/*.csv</code>                              |
| Retrieval + processing (MAXI)  | label freeze, Sesame matching, 1-day light curves, cuts; $N_{\text{H}}$ | 37, 38, 45                     | <code>data/v7c/maxi_points.csv</code> , <code>data/v8d/</code>        |
| Bursts and states              | Standard-1 burst detector, MINBAR match, rms state classes              | 17, 31–34, 42                  | <code>results/v4/burst_*</code> , <code>results/v7a/states.csv</code> |

| Stage       | What happens                                        | Scripts                                            | Main outputs                                                      |
|-------------|-----------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------|
| Data checks | integrity, response variation, XSPEC cross-check    | 06, 10                                             | <code>results/data_checks.md</code>                               |
| Modelling   | LOSO, nested selection, frozen models               | 07, 11, 16, 27, 30, 44, 49, 54, 57, 59, 63         | OOF prediction tables                                             |
| Inference   | source bootstrap, permutation tests, conformal sets | 09 and analysis scripts; <code>v4lib-v10lib</code> | <code>results/*/bootstrap*.csv</code> ,<br><code>*_nuc.csv</code> |
| Summaries   | cross-round figures and tables                      | 15, 40, 46, 51, 64                                 | <code>figures/final/</code> ,<br><code>results/final_*</code>     |

### 1.3 Analysis rounds and how a round runs

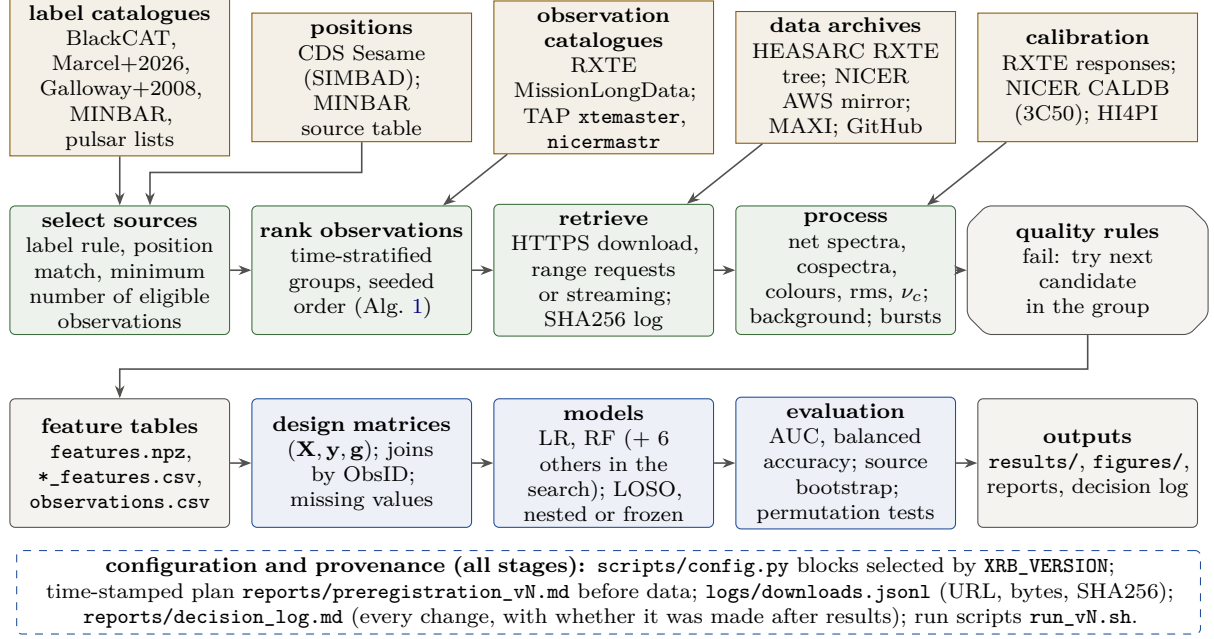
The project grew in numbered rounds (v1–v14). Table 2 lists them briefly. Every round from v2 on follows the same procedure (Figure 2).

**Table 2:** Analysis rounds (repository versions). “Data” is what the round added; plans are `reports/preregistration_vN.md`.

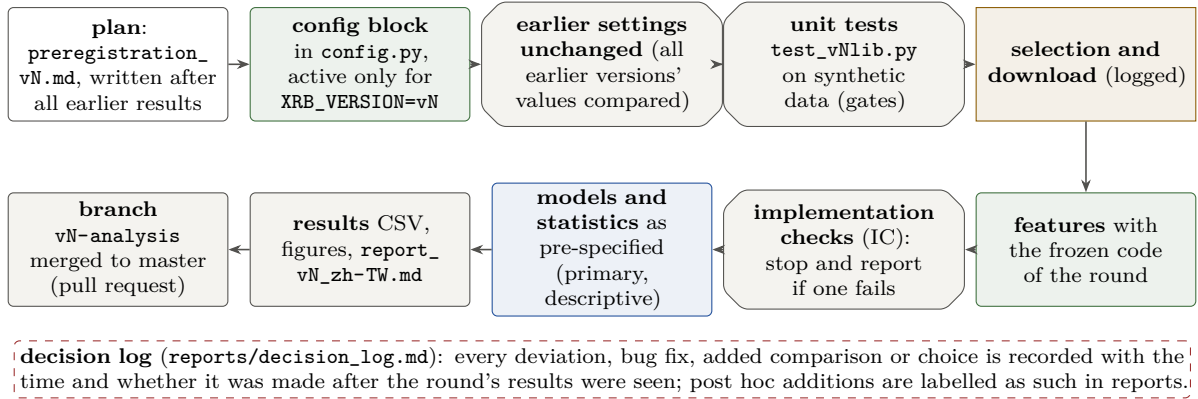
| Round  | Data added                                                           | Main method added                                                                                     |
|--------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| v1     | RXTE Standard-2 spectra, 8 sources                                   | pipeline, LOSO, LR/RF on representations A and B                                                      |
| v2     | 31 sources, 456 observations; Standard-1 dead time                   | two-colour baseline H, source bootstrap, XSPEC check                                                  |
| v3     | 3–25 keV grid; 14 BH candidates                                      | nested increment tests (H+B vs H), source-level AUC                                                   |
| v4     | HEXTE; earlier gain epochs (v4b2); all 7949 epoch-5 pointings (v4b1) | 8-algorithm nested search, burst detector, source-equal weights                                       |
| v5–v6  | Standard-1 timing features; 23 external sources                      | cospectral rms $T_1$ – $T_3$ ; all-pairs estimator; $\nu_c$ ; permutation test; CCTLR; conformal sets |
| v7     | MINBAR; Cyg X-1, LMC X-1, LMC X-3; MAXI                              | rms state classes; burst matching; MAXI pipeline and reproduction of de Beurs et al.                  |
| v8     | GS 1354–64, SS 433; RXTE event mode; HI4PI                           | state-stratified bootstrap; high-frequency cospectra; $N_H$ correction                                |
| v9–v10 | NICER N1 (prefix, then full files)                                   | NICER streaming parser and features; pooled RXTE+NICER tests                                          |
| v11    | NICER N2 (unseen observations)                                       | frozen out-of-source prediction; Lorentzian fits                                                      |
| v12    | 3C50 background for N1+N2                                            | background correction and screening                                                                   |
| v13    | NICER N3 (unseen observations)                                       | everything frozen before download                                                                     |
| v14    | –                                                                    | cross-round summary only                                                                              |

### 1.4 Repository layout and conventions

- **Version switch.** Every script sets or reads the environment variable `XRB_VERSION`, and `scripts/config.py` activates the corresponding parameter block. Later versions extend earlier blocks, and each new block is checked not to change the values of earlier versions. `scripts/common.py` maps the version to the output folders: `data/<v>/`, `results/<v>/`, `figures/<v>/` and `logs/<v>/` (v1 uses the top-level folders).
- **Raw data.** Raw data stay out of git. RXTE products of v1–v3 and the v8 event files are in `data/raw/`. Large sets live under `XRB_EXTERNAL_RAW` (default `~/xrb-pilot-data/raw`): v4b1, the NICER prefixes and tails, the v13 cleaned files and the 3C50 spectra (`~/xrb-pilot-data/nicer_bkg_spectra`). The NICER CALDB is under `~/xrb-pilot-data/caldb`.
- **In git.** Everything derived is committed: source and candidate tables, every tried observation with its status and exclusion reason, feature tables, out-of-fold predictions, result tables, figures and logs.



**Figure 1:** End-to-end workflow. Amber: external databases; green: selection, retrieval and processing; blue: data model, models and evaluation; grey: stored outputs; chamfered: a gate that can reject an observation. An observation that fails the quality rules is replaced by the next ranked candidate of the same time group, so the sample is defined before any spectrum or light curve is examined.



**Figure 2:** Life cycle of one analysis round. The plan and the configuration are committed before the round's data are downloaded. The implementation checks and unit tests are written before real results exist; if a check fails, the round stops and the failure is reported rather than the rule being changed.

- **Randomness.** The global seed is `SEED = 42`. Sampling seeds are offsets of it (Table 8); the bootstrap, permutations and models use it directly.
- **Shared modules.** `common.py` (download and progress log), `spectra.py` (Standard-2 reader without HEASoft), `config.py` (settings) and the version libraries `v3lib.py`–`v11lib.py`. Each version library is tested by a `test_vNlib.py`.

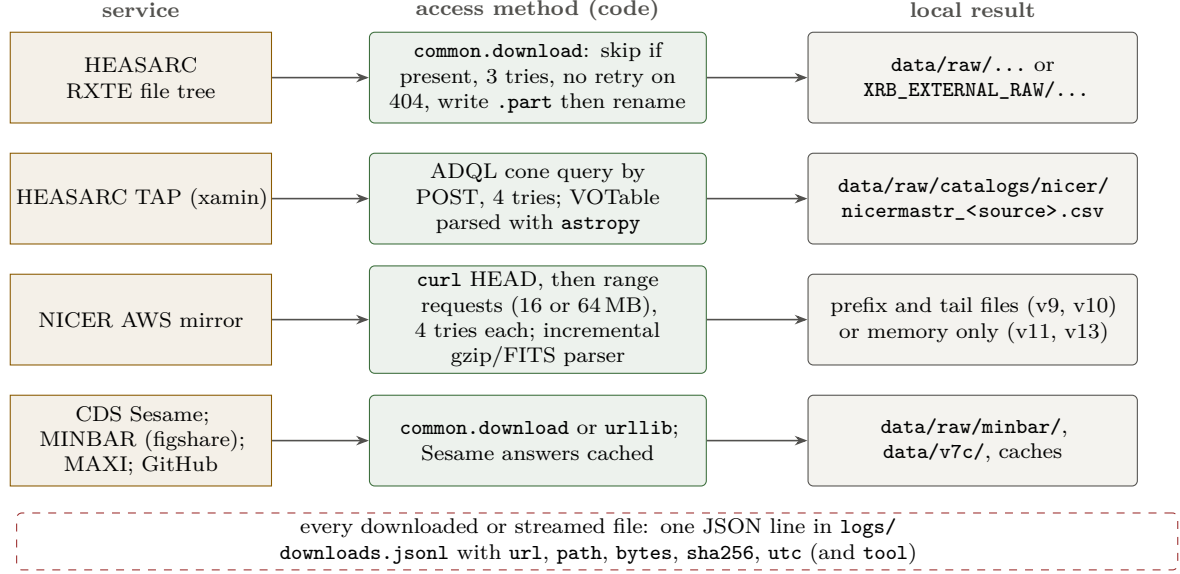
## 2 External databases and how the data are fetched

All inputs come from public services and are obtained by the scripts, with no manual downloads except the NICER CALDB archive (Section 2.3). Table 3 lists every external database, what is taken from it and how. Figure 3 shows how requests flow from the scripts to local storage and to the download log.

**Table 3:** External databases used by the workflow. “Records” and volumes are from `logs/downloads.jsonl` (Section 2.5).

| Database                                  | What is taken                                                                                                                                                                     | Access                                                                                                                                                                      | Code                                                                                                  |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <i>Labels and positions</i>               |                                                                                                                                                                                   |                                                                                                                                                                             |                                                                                                       |
| BlackCAT<br>(Corral-Santana et al., 2016) | dynamically confirmed BHs (v1–v4); candidate list (v3c, descriptive)                                                                                                              | names transcribed into <code>config.py</code> ; candidates checked against the BlackCAT web pages                                                                           | <code>config.py</code> (v2, v3c blocks)                                                               |
| Marcel et al. (2026), Table 1             | 25 dynamically confirmed BHs (v7 onwards)                                                                                                                                         | names transcribed into the scripts                                                                                                                                          | <code>38_v7c2_select.py</code> , <code>47_v9_select.py</code>                                         |
| Galloway et al. (2008)                    | type-I bursters, appendix A (v1–v3), Table 3 (v4b2)                                                                                                                               | names transcribed into <code>config.py</code>                                                                                                                               | <code>config.py</code> , <code>20_v4b2_select.py</code>                                               |
| MINBAR DR1<br>(Galloway et al., 2020)     | burst list, observation list, source table with positions                                                                                                                         | figshare files 23201936, 24131849, 22851017 (2.5 MB, 51.9 MB, 0.7 MB), MD5 checked                                                                                          | <code>31_v7b1_minbar.py</code> , <code>v7lib.load_minbar</code>                                       |
| Pulsar lists                              | AMXPs and slow pulsars (Patruno & Watts 2021); HMXB/LMXB catalogues of Liu et al.                                                                                                 | names transcribed; catalogue tables from CDS (8 files)                                                                                                                      | <code>28_v6_select.py</code> , <code>38_v7c2_select.py</code>                                         |
| CDS Sesame<br>(SIMBAD)                    | RA/Dec of every named source                                                                                                                                                      | GET <code>https://cds.unistra.fr/cgi-bin/nph-sesame/-oI/SNV?&lt;name&gt;</code> ; the %J line is parsed; results cached in CSV                                              | <code>sesame()</code> in 20, 28, 38                                                                   |
| <i>Observation catalogues</i>             |                                                                                                                                                                                   |                                                                                                                                                                             |                                                                                                       |
| RXTE<br>MissionLongData                   | per-source FITS observation list: <code>OBSID</code> , <code>TIME</code> , <code>EXPOSURE</code> , <code>STD1RATE</code> , <code>RA</code> , <code>DEC</code> , <code>NPCU</code> | directory listing and <code>&lt;name&gt;.fits.gz</code> under <code>.../xte/data/archive/MissionLongData/</code>                                                            | 02, 03, 20, 35, 41                                                                                    |
| HEASARC TAP<br><code>xtemaster</code>     | RXTE pointings near a position (sources without a MissionLongData file)                                                                                                           | ADQL cone query, HTTP POST to <code>.../xamin/vo/tap/sync</code>                                                                                                            | <code>41_v8b_select.py</code>                                                                         |
| HEASARC TAP<br><code>nicermastr</code>    | NICER observations near a position: <code>obsid</code> , <code>time</code> , <code>exposure</code> , <code>num_fpm</code>                                                         | same service; one query per source, cached per source                                                                                                                       | <code>47_v9_select.py</code>                                                                          |
| <i>Data archives</i>                      |                                                                                                                                                                                   |                                                                                                                                                                             |                                                                                                       |
| HEASARC RXTE<br>archive                   | Standard-2 Standard Products, Standard-1 light curves, event files, HEXTE products                                                                                                | HTTPS file tree; Python <code>urllib</code> (event files: <code>curl</code> ); 46 731 records, 5.2 GB                                                                       | <code>common.download</code> , <code>v8lib.download_curl</code>                                       |
| HEASARC NICER<br>(AWS open-data mirror)   | cleaned ( <code>_c1</code> ) and unfiltered ( <code>_ufa</code> ) event files                                                                                                     | <code>curl</code> HTTP range requests; 3 384 records, 229.5 GB read                                                                                                         | <code>v9lib.py</code> , <code>v10lib.py</code> , <code>v11lib.py</code> , <code>v12_bkg3c50.sh</code> |
| MAXI/GSC                                  | source index; 1-day light curves (2–4, 4–10, 10–20 keV)                                                                                                                           | 13 category pages <code>http://maxi.riken.jp/top/lc_&lt;cat&gt;.html</code> ; <code>http://maxi.riken.jp/star_data/&lt;J&gt;/&lt;J&gt;_g_lc_1day_all.dat</code> (143 files) | <code>38_v7c2_select.py</code>                                                                        |
| GitHub (de Beurs et al.)                  | processed MAXI data and code of de Beurs et al. (2022)                                                                                                                            | GitHub API for the commit and file tree, then raw files at commit 488b785; 487 files                                                                                        | <code>37_v7c1_repro.py</code>                                                                         |
| <i>Calibration</i>                        |                                                                                                                                                                                   |                                                                                                                                                                             |                                                                                                       |
| RXTE responses<br>NICER CALDB             | <code>xp&lt;obs&gt;.rsp</code> per observation<br>3C50 background libraries                                                                                                       | with the Standard Products <code>goodfiles_nicer_xti.tar.gz</code> (112 180 364 bytes, SHA256 recorded in the decision log)                                                 | <code>spectra.read_rsp</code><br><code>v12_bkg3c50.sh</code>                                          |

| Database | What is taken                           | Access                                     | Code                        |
|----------|-----------------------------------------|--------------------------------------------|-----------------------------|
| HI4PI    | Galactic $N_H$ at MAXI source positions | HEASoft <code>nh</code> with its HI4PI map | <code>v8d_heasoft.sh</code> |



**Figure 3:** Data acquisition. Each service is reached through one access function. Downloads are idempotent: a file that already exists is not fetched again, and every new file is logged with its checksum.

## 2.1 Label lists and positions

Labels are attached to *sources*, never to observations. The label sources are published tables, and the realised source lists are the `data/*/sources.csv` files:

- **BHs, v1–v4:** the dynamically confirmed systems of BlackCAT Table 4, transcribed into the v2 block of `config.py` with their MissionLongData catalogue names.
- **BHs, v7 onwards:** the 25 systems of Marcel et al. (2026) Table 1, written as a list in the selection scripts.
- **NSs:** type-I bursters of Galloway et al. (2008) (appendix A sections cited per source in `config.py`). From v7 the MINBAR source table (115 bursters) is used instead; the external RXTE set adds accretion-powered pulsars.
- **BH candidates** (BlackCAT, X-ray evidence only) are never labels. In v3c they were scored as prediction targets, after the list had been checked against the BlackCAT web pages and frozen (`reports/v3c_candidate_verification.md`).

Positions come from three places, depending on the round:

- **v1–v3:** the median pointing in the source’s RXTE catalogue.
- **v4b2 onwards:** CDS Sesame. The script requests `https://cds.unistra.fr/cgi-bin/nph-sesame/-oI/SNV?<name>`, reads RA and Dec from the line starting with %J, and caches the answer (`data/v7c/sesame_positions.csv`, `data/v4b2/missionlongdata_positions.csv`).
- **NSs from v7:** the MINBAR source table (`minbar_sources_v2.7.fits`, columns NAME, RA\_OBJ, DEC\_OBJ).

The MINBAR burst table (`minbar.txt`, machine-readable format) supplies burst start times and durations (`v7lib.load_minbar`). The three MINBAR files are downloaded from figshare; the MD5 sums of the two burst-related files are fixed in `V7_MINBAR_FILES` in `config.py`.

## 2.2 RXTE: catalogues and archive files

**Observation catalogues.** The archive root is <https://heasarc.gsfc.nasa.gov/FTP/xte/data/archive/>. Its `MissionLongData/` directory holds one FITS table per catalogued target. `02_catalogs.py` and `03_select_observations.py` download `MissionLongData/<name>.fits.gz` and read extension 1 with `astropy`. The observation time is `TIME + MJDREFI + MJDREFF` (MJD), and duplicated OBSIDs are dropped. For v4b2, the whole directory listing (291 catalogues) was downloaded and each catalogue’s median pointing was matched to Sesame positions within  $0.1^\circ$ . For new BHs without a catalogue file (v8b), the HEASARC TAP service is queried instead:

```
SELECT obsid, target_name, ra, dec, time, exposure FROM xtemaster
WHERE CONTAINS(POINT('ICRS',ra,dec),CIRCLE('ICRS',<ra>,<dec>,0.1))=1
```

The query is sent by HTTP POST to <https://heasarc.gsfc.nasa.gov/xamin/vo/tap/sync> with `REQUEST=doQuery`, `LANG=ADQL`.

**Archive layout and file names.** An ObsID PPPPP-TT-VV-SS lives under `A0<n>/P<PPPPP>/<obsid>/`. The files used are listed below.

**Table 4:** RXTE files read per observation (o = ObsID without hyphens).

| Path below the ObsID directory                             | Content                                                           | Used in                       |
|------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------|
| <code>stdprod/xp&lt;o&gt;_s2.pha.gz</code>                 | PCA Standard-2 total (source + background) spectrum, 129 channels | v1–v8                         |
| <code>stdprod/xp&lt;o&gt;_b2.pha.gz</code>                 | pcabackest model background spectrum, same PCUs and GTIs          | v1–v8                         |
| <code>stdprod/xp&lt;o&gt;.rsp.gz</code>                    | response (RMF×ARF) for that PCU combination, with EBOUNDS         | v1–v8                         |
| <code>pca/FS46_*.gz</code>                                 | Standard-1: 0.125s counts per PCU (XeCntPcu0–4) and array rates   | dead time (v2+), timing (v5+) |
| <code>pca/SE*.evt.gz</code>                                | event-mode files (TIME, PCUID)                                    | v8c                           |
| <code>stdprod/xh&lt;o&gt;_s1.pha, _b1.pha + RMF/ARF</code> | HEXTE cluster-B spectra and responses                             | v4b3                          |

**Locating the AO directory.** The AO number is not stored in the catalogue. `04_fetch_process.py` guesses it from the proposal prefix (5→AO5, ..., 90→AO9, 91→AO10, ..., 96→AO15). If the guess returns HTTP 404, it tries AO4–AO16 in turn and caches the AO that worked for the proposal. (The v4b2 copy of the script also scans AO1–AO4 for the early gain epochs.) The Standard-1 file names are taken from the HTML index of the `pca/` directory (regular expression `FS46_[^"]+\.``gz`).

**Download function.** `common.download(url, relative_path)` does the following:

1. returns the local file if it exists (re-runs never re-download);
2. otherwise fetches the URL with `urllib` (45s timeout), retrying up to three times with a 2s pause, but not on HTTP 404;
3. writes the data to `<name>.part` and renames it, so an interrupted run leaves no truncated files;
4. appends a record with `url`, `path`, `bytes`, `sha256` and `utc` to `logs/downloads.jsonl`, under a lock because downloads run in a thread pool (4 workers in `04_fetch_process.py`).

The RXTE event files of v8c (2.4 GB) were fetched with `v8lib.download_curl`, which streams with `curl` into a `.part` file and writes the same log record.

## 2.3 NICER: TAP query, mirror and range requests

**Observation list.** For every source, `47_v9_select.py` queries the HEASARC TAP service:

```
SELECT obsid, name, ra, dec, time, exposure, num_fpm FROM nicermastr
WHERE CONTAINS(POINT('ICRS',ra,dec),CIRCLE('ICRS',<ra>,<dec>,0.05))=1
```

The query runs with 6 parallel requests and 4 tries each. The VOTable is saved per source as `data/raw/catalogs/nicer/nicermastr_<source>.csv`. The later rounds (v11, v13) select from these cached tables, so all three NICER observation sets come from the same catalogue snapshot.

**File location.** Cleaned event files are read from HEASARC’s AWS open-data mirror, which is byte-identical to the HEASARC site (checked on the first 4 MB) and about ten times faster from the analysis machine:

```
https://nasa-heasarc.s3.amazonaws.com/nicer/data/obs/<YYYY_MM>/<obsid>/xti/event_
cl/ni<obsid>_0mpu7_cl.evt.gz
```

YYYY\_MM is the month of the catalogue start time. If the file is not found (HTTP 403 or 404 on the mirror), the previous and next months are tried. The unfiltered file `..._0mpu7_ufa.evt.gz` is fetched only for the 3C50 background (Section 5.2).

**Range requests and the streaming parser.** The files are gzip-compressed FITS tables of up to several GB, so they are never downloaded blindly:

1. A HEAD request (`curl -I`) checks that the file exists and gives its size.
2. The file is fetched in HTTP range requests (`curl -r a-b`; options `--connect-timeout 30`, `--speed-limit 20000`, `--speed-time 60`), each with up to 4 tries and a 5 s pause.
3. Each piece is passed to an incremental parser (`v9lib.PrefixParser`). The parser decompresses the gzip stream with `zlib`, reads the primary and `EVENTS` headers from 2880-byte blocks, and converts complete binary-table rows to arrays of `TIME`, `PI` and `DET_ID` using a numpy dtype built from the `TFORMn` keywords.

No HEASoft is needed to read the events. The three reading modes are listed in Table 5. Parsing was checked to agree with `astropy` and across the modes (implementation checks of v9–v11).

**Table 5:** NICER reading modes.

| Round             | Function                          | Request size         | Stop rule                                                | Storage                                                                                                          |
|-------------------|-----------------------------------|----------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| v9 (prefix)       | <code>v9lib.stream_prefix</code>  | 16 MB                | $\geq 6$ gap-free 128 s segments, end of file, or 300 MB | compressed prefix + JSON metadata in <code>XRB_EXTERNAL_RAW/nicer/&lt;obsid&gt;/</code>                          |
| v10 (tail)        | <code>v10lib.download_tail</code> | 64 MB                | end of file or 1 GB compressed                           | tail in <code>nicer_tail/&lt;obsid&gt;/</code> ; prefix + tail parsed as one stream                              |
| v11, v13 (stream) | <code>v11lib.stream_events</code> | 64 MB, 6 in parallel | end of file or 1 GB                                      | nothing stored (SHA256 of the bytes read is logged); v13 keeps cleaned files of faint observations for one check |

**3C50 inputs and the CALDB.** For the background model, the script `v12_bkg3c50.sh` runs inside WSL Ubuntu with HEASoft 6.37.1. It:

1. downloads the cleaned and unfiltered event files with `curl` (3 tries) and prints their SHA256 and size, which the Python driver writes to the download log;
2. unzips them and runs `nibackgen3C50` (Section 5.2);
3. keeps only the resulting total and background spectra (`tot_<obsid>.pi`, `bkg_<obsid>.pi`);
4. deletes the event files.

The NICER CALDB (`goodfiles_nicer_xti.tar.gz` with `caldb.config` and `alias_config.fits`) was downloaded once before the v12 data and unpacked under `~/xrb-pilot-data/caldb`; its size and SHA256 are in the decision log.

## 2.4 MAXI, published code and absorption

- **MAXI matching.** The label lists are frozen first. The MAXI source index is read from the 13 category pages ([http://maxi.riken.jp/top/lc\\_<cat>.html](http://maxi.riken.jp/top/lc_<cat>.html)). MAXI names are resolved with Sesame and matched to the label sources within  $0.1^\circ$ , with a  $0.3^\circ$  fallback to the position encoded in the MAXI J-name. MAXI sources whose name contains “with” (blends) are excluded, as are ambiguous matches (Section 5.3).
- **MAXI light curves.** The 1-day light curves are downloaded from [http://maxi.riken.jp/star\\_data/<J>/<J>\\_g\\_lc\\_1day\\_all.dat](http://maxi.riken.jp/star_data/<J>/<J>_g_lc_1day_all.dat) with `common.download`.
- **Published code.** For the reproduction of [de Beurs et al. \(2022\)](#), `37_v7c1_repro.py` asks the GitHub API for the current commit and file tree of `zdebeurs/3ML_methods_for_XRB_classification`. It then downloads every file from `raw.githubusercontent.com` at that commit (488b785).
- **Absorption.** For the absorption correction, `v8d_heasoft.sh` runs HEASoft `nh` (HI4PI, average within  $0.1^\circ$ ) at each MAXI source position. It also runs XSPEC to tabulate `tbabs` × power-law transmissions (Section 5.3).

## 2.5 Provenance log and verification

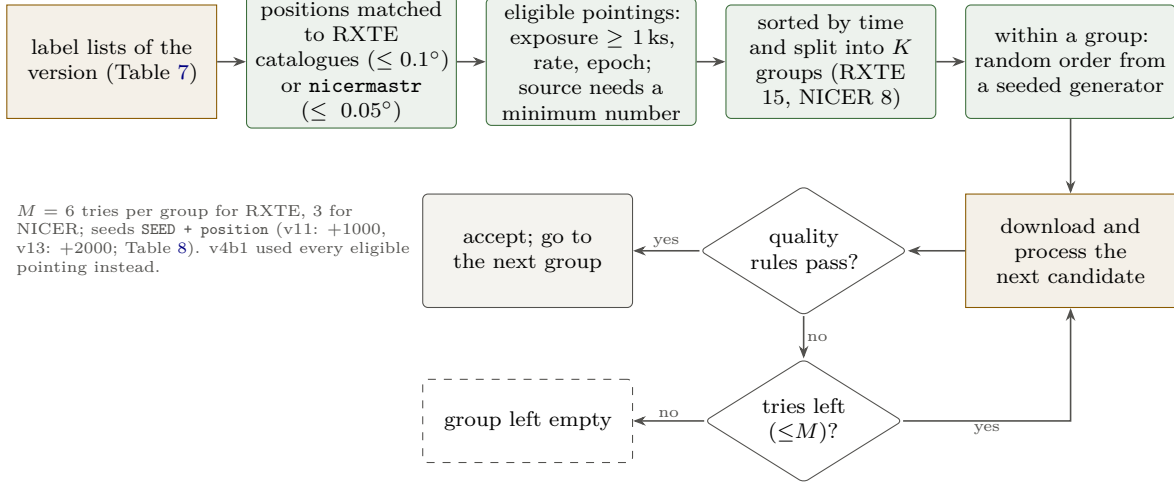
`logs/downloads.jsonl` has one JSON object per downloaded or streamed file. Its fields are `url`, `path` (or a note such as “deleted after 3C50”), `bytes`, `sha256` and `utc`; the NICER functions add `tool`, `file_bytes`, `stop`, `rows` and `offsets`. At commit 666c2e0 it holds 50 773 records (Table 6). Every input can be verified byte for byte after re-download. This was done once in full: on a second machine (macOS), the 2 689 v1–v2 raw files were re-downloaded from the logged URLs and all SHA256 values matched. The rebuilt `features.npz` agreed with the original to  $\leq 10^{-16}$ .

**Table 6:** Download log by host (`logs/downloads.jsonl`, commit 666c2e0). Streamed NICER records count the bytes read, although the bytes were not stored.

| Host                                       | Records | Volume    | Tool(s)                                                                                  |
|--------------------------------------------|---------|-----------|------------------------------------------------------------------------------------------|
| <code>heasarc.gsfc.nasa.gov</code>         | 46 731  | 5.2 GB    | <code>urllib</code> ( <code>common.download</code> ); <code>curl</code> for event files  |
| <code>nasa-heasarc.s3.amazonaws.com</code> | 3 384   | 229.5 GB  | <code>curl</code> range requests (prefix, tail, stream); <code>curl</code> in WSL (3C50) |
| <code>raw.githubusercontent.com</code>     | 487     | 0.01 GB   | <code>urllib</code>                                                                      |
| <code>maxi.riken.jp</code>                 | 156     | 0.06 GB   | <code>urllib</code>                                                                      |
| <code>cdsarc.cds.unistra.fr</code>         | 8       | < 0.01 GB | <code>urllib</code>                                                                      |
| <code>arxiv.org</code>                     | 4       | < 0.01 GB | <code>urllib</code>                                                                      |
| <code>ndownloader.figshare.com</code>      | 3       | 0.06 GB   | <code>urllib</code>                                                                      |
| total                                      | 50 773  | 234.8 GB  |                                                                                          |

### 3 Sample construction: sources and observations

Sample construction runs in two stages. First, *sources* are chosen and labelled from published tables. Second, for each source, *observations* are chosen from the archive catalogue by a rule that uses only times, exposures, count rates and positions, never the spectrum or the light curve. Data are downloaded and processed only after both lists are written. The processing can reject an observation (quality rules); the next candidate of the same time group then takes its place. Figure 4 summarises the procedure.



**Figure 4:** Selection of sources and observations. Apart from the quality rules, the choice of an observation depends only on the label list, the catalogue position, the eligibility rules, the time order and a seeded random order.

#### 3.1 Source labels

The label is a property of the source:  $y = 1$  for a black hole and  $y = 0$  for a neutron star. A BH needs a dynamical mass function, and an NS needs type-I X-ray bursts or coherent accretion-powered pulsations. BH candidates (X-ray evidence only) are never labels. Table 7 lists the rule of each round. The realised source lists, with the evidence and a reference URL for every source, are `data/<v>/sources.csv`.

**Confused fields.** NICER does not image, so a MINBAR source with another MINBAR source within  $0.05^\circ$  is flagged `excluded_confused` and not used. Nine of the 115 MINBAR sources are removed this way in `47_v9_select.py`. For RXTE, the  $0.1^\circ$  pointing rule plays the same role; `config.py` notes, for example, that 4U 1907+097 lies about  $0.4^\circ$  from the candidate XTE J1908+094.

#### 3.2 Catalogue-level eligibility

For RXTE (`03_select_observations.py` and its copies), each source’s MissionLongData catalogue goes through the steps below. The number of rows left after each step is written to `results/<v>/attrition_catalog.csv`:

1. drop duplicated OBSIDs;
2. keep the time window of the round: gain epoch 5 (MJD 51677 to 54094 in v1, to 55931 from v2), or all gain epochs (v4b2, v6, v8b);
3. keep  $\text{EXPOSURE} \geq 1000 \text{ s}$ ;
4. keep  $\text{STD1RATE} \geq 5 \text{ c s}^{-1} \text{ PCU}^{-1}$ ;
5. keep pointings within  $0.1^\circ$  of the source position (great-circle distance);

**Table 7:** Labelling and eligibility rules by round. “Pointing” is the largest separation allowed between an observation’s pointing and the source position. Every round also requires a catalogue exposure  $\geq 1000$  s (RXTE: and  $\text{STD1RATE} \geq 5 \text{ cs}^{-1} \text{ PCU}^{-1}$ ).

| Round          | BH rule                                                    | NS rule                                                                 | Pointing; time window; minimum eligible                          |
|----------------|------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------|
| v1             | 4 BlackCAT dynamical BHs                                   | 4 Galloway+2008 bursters                                                | 0.1° from the catalogue median; gain epoch 5 to MJD 54094; 30    |
| v2, v3         | every BlackCAT Table 4 BH with a MissionLongData catalogue | every Galloway+2008 appendix-A burster with a catalogue                 | 0.1°; gain epoch 5 (MJD 51677–55931); 10                         |
| v3c            | 16 BlackCAT candidates (prediction targets only)           | –                                                                       | as v2                                                            |
| v4b2           | BlackCAT BHs found by position                             | Galloway+2008 Table 3 bursters found by position                        | 0.1° from Sesame; all gain epochs; 10                            |
| v6             | HMXB BHs (Cyg X-1, LMC X-3)                                | accreting millisecond pulsars; 4 slow pulsars as a stress set           | as v4b2                                                          |
| v7b2           | Cyg X-1, LMC X-1, LMC X-3 (Marcel et al., 2026)            | –                                                                       | 0.1° from Sesame; gain epoch 5; 10                               |
| v7c (MAXI)     | Marcel et al. (2026) Table 1                               | MINBAR bursters without pulsations; pulsars as a separate class         | MAXI source within 0.1° (0.3° fallback); $\geq 100$ daily points |
| v8b            | Marcel+2026 BHs not used before (GS 1354–64, SS 433)       | –                                                                       | 0.1° from Sesame; all gain epochs; 3                             |
| v9–v13 (NICER) | Marcel+2026 Table 1, Sesame positions                      | MINBAR bursters; entries with another MINBAR entry within 0.05° removed | target within 0.05°; mission; 5                                  |

- drop the source if fewer than the minimum number of pointings remain (Table 7). The source stays in `sources.csv` with `included = False`.

For NICER (47\_v9\_select.py), the TAP query already restricts the target position to 0.05°. The remaining steps are: drop rows without a time, drop duplicated ObsIDs, keep `exposure`  $\geq 1000$  s, and require at least 5 eligible observations and a non-confused field.

### 3.3 Time-stratified ranking

The eligible observations of a source are sorted by time and split into  $K$  consecutive groups of nearly equal size (`numpy.array_split`):  $K = 15$  for RXTE, 8 for NICER. A seeded generator puts the observations of each group in a random order (Algorithm 1). The seed is `SEED` plus an offset plus the 1-based position of the source in the source list (Table 8). The result is `data/<v>/observation_candidates.csv`, with one row per eligible observation, its group (`time_bin`) and its rank in the group (`rank_in_bin`). This file is written before any data file is downloaded.

The stratification spreads each source’s sample over its history, across outbursts and states. Because the order is random within a group, the choice does not depend on the data. If a source has fewer eligible observations than groups, some groups are empty and every observation is a candidate.

### 3.4 Acceptance loop and quality rules

The fetch scripts work through the (source, group) pairs in parallel: 4 threads for RXTE; for NICER, 8 threads in v9 and 2 in v11 and v13. Within a pair the candidates are tried in rank order. Every tried candidate becomes one row of `data/<v>/observations.csv`, with its status, its reason, the download URL and the quality quantities. Rejected observations are therefore documented, not lost.

Two further screens act after acceptance and do not trigger a replacement. Type-I bursts are flagged, or the affected segments removed (Section 4.9). From v12, NICER observations are screened

---

**Algorithm 1** Time-stratified ranking and acceptance. RXTE: 03\_select\_observations.py with 04\_fetch\_process.py; NICER: 47/55/60 with 48/52/56/61.

---

**Require:** eligible observations  $\{(o_i, t_i)\}$  of one source; number of groups  $K$ ; seed  $\sigma$ ; tries per group  $M$

- 1: sort by  $t_i$  and split the index list into  $K$  consecutive groups  $G_1, \dots, G_K$
- 2:  $\text{rng} \leftarrow \text{default\_rng}(\sigma)$
- 3: **for**  $b = 1, \dots, K$  **do**
- 4:    $\pi \leftarrow \text{rng.permutation}(G_b)$   $\triangleright$  rank within the group; written to the candidate table
- 5: **end for**
- later, in the fetch script, for each group independently —
- 6: **for**  $r = 1, \dots, \min(M, |G_b|)$  **do**
- 7:   download and process  $o_{\pi_r}$ ; record status and reason
- 8:   **if** every quality rule passes **then** accept  $o_{\pi_r}$ ; **break**
- 9:   **end if**
- 10: **end for**

---

**Table 8:** Sampling parameters. “Position” is the 1-based position of the source in the list processed by the script (for NICER, data/v9/sources.csv, counting excluded sources too). **SEED** = 42.

| Sample                      | Groups $K$   | Seed $\sigma$          | Tries per group $M$ |
|-----------------------------|--------------|------------------------|---------------------|
| RXTE v1, v2, v3c, v7b2, v8b | 15           | SEED + position        | 6                   |
| RXTE v4b2 (new sources)     | 15           | SEED + 100 + i         | 6                   |
| RXTE v4b1                   | all eligible | —                      | 1                   |
| NICER v9 (N1)               | 8            | SEED + position        | 3                   |
| NICER v11 (N2)              | 8            | SEED + 1000 + position | 3                   |
| NICER v13 (N3)              | 8            | SEED + 2000 + position | 3                   |

**Table 9:** Post-download quality rules. An observation is accepted only if no rule fires. “Reason” is the string written to observations.csv.

| Data      | Reason                            | Rule                                                           |
|-----------|-----------------------------------|----------------------------------------------------------------|
| RXTE PCA  | fetch_or_read_error               | a file is missing in every AO directory, or cannot be read     |
|           | nonfinite                         | a rebinned rate or error is not finite                         |
|           | short_exposure                    | Standard-2 EXPOSURE < 1000 s                                   |
|           | low_snr                           | net 5–25 keV rate / its error < 10 (Equation (4))              |
|           | high_deadtime                     | dead-time fraction > 0.10 (Equation (2))                       |
|           | coord_mismatch                    | header pointing > 0.1° from the source position                |
|           | nonstandard_channels              | spectrum or response does not have 129 channels                |
|           | ebounds_coverage                  | the response’s energy range does not cover the common grid     |
| NICER XTI | error: ...                        | download or parsing error (after retries)                      |
|           | missing (event file not found)    | no file in the catalogue month or the months before and after  |
|           | missing (< 3 valid segments)      | fewer than 3 usable 128 s segments (Section 5.1)               |
|           | excluded (2–10 keV rate x < 20.0) | mean 2–10 keV rate in the valid segments < 20 cs <sup>−1</sup> |

on their 3C50 background fraction (Section 5.2).

### 3.5 Disjoint NICER observation sets

NICER was used three times, each time with observations that had never been tried before:

- **N1** (v9, re-read in full in v10): the first time-stratified sample of the v9 sources.
- **N2** (v11): for each included v9 source, the eligible observations *not* among the 792 v9 tries, ranked with `SEED + 1000 + position`. Sources not included in v9 but with 1–4 eligible observations and a non-confused field contribute every eligible observation, one per group.
- **N3** (v13): eligible observations of the included v9 sources that are among neither the 792 v9 tries nor the 588 v11 tries, ranked with `SEED + 2000 + position`.

The selection scripts compute the overlap with the earlier tries and stop with exit code 3 if it is not empty (check IC3 in v11). All three sets come from the same cached `nicermastr` tables, so no new catalogue query can change the candidates.

### 3.6 Resulting samples

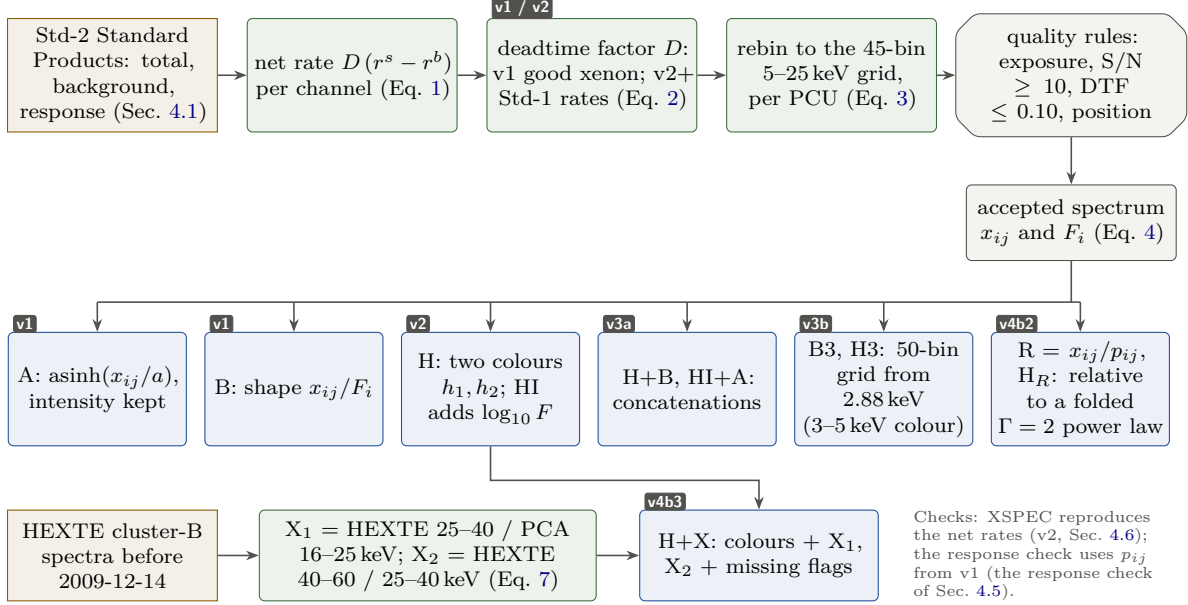
Table 10 lists the samples that enter models or tests. Counts are recomputed from the committed `observations.csv` and processed tables.

**Table 10:** Samples. “Tried” counts every downloaded candidate; “Obs.” counts the accepted (or kept) observations; “Sources” gives the total and the BH/NS split.

| Sample                    | Round,<br>instrument  | Tried | Obs.    | Sources                     | Use                                                      |
|---------------------------|-----------------------|-------|---------|-----------------------------|----------------------------------------------------------|
| pilot                     | v1, PCA               | 120   | 120     | 8 (4/4)                     | pipeline test                                            |
| S1 (development)          | v2, PCA, epoch 5      | 462   | 456     | 31 (7/24)                   | training and LOSO in<br>v2–v8, v10                       |
| BH candidates             | v3c, PCA              | 210   | 210     | 14 (14/0)                   | prediction only                                          |
| S3 (all pointings)        | v4b1, PCA, epoch 5    | 7 975 | 7 949   | 31 (7/24)                   | observation-budget checks                                |
| v4b2 sources              | v4b2, PCA, all epochs | 221   | 221     | 15 (2/13)                   | S2 = S1 $\cup$ v4b2 (46 sources, 677 obs.)               |
| E <sub>new</sub> + stress | v6, PCA, all epochs   | 180   | 180     | 12 (2/10)                   | 8 new sources (120 obs.); 4 slow pulsars (60 obs.)       |
| E (external)              | v6                    | –     | 341     | 23 (4/19)                   | frozen-model test                                        |
| persistent BHs            | v7b2, PCA, epoch 5    | 45    | 45      | 3 (3/0)                     | Cyg X-1, LMC X-1, LMC X-3                                |
| new BHs                   | v8b, PCA, all epochs  | 26    | 23      | 2 (2/0)                     | GS 1354–64, SS 433                                       |
| X (external)              | v8b                   | –     | 379     | 26 (7/19)                   | frozen-model test; 99 hard-like obs. (5/15 sources)      |
| MAXI                      | v7c2, GSC             | –     | 132 057 | 100: 13 BH, 48 NPNS, 39 PSR | daily points (not observations); point-level classifiers |
| N1                        | v9 (prefix), XTI      | 792   | 368     | 63 (9/54)                   | first NICER sample                                       |
| N1 full files             | v10, XTI              | 368   | 366     | 63 (9/54)                   | N1 re-read to 1 GB                                       |
| N2                        | v11, XTI              | 588   | 318     | 60 (9/51)                   | unseen observations                                      |
| N1+N2, 3C50               | v12, XTI              | –     | 534     | 58 (9/49)                   | corrected and screened; frozen training set for v13      |
| N3                        | v13, XTI              | 412   | 238     | 37 (8/29)                   | unseen observations                                      |
| N3, 3C50                  | v13, XTI              | –     | 198     | 35 (8/27)                   | corrected and screened test set                          |

## 4 RXTE processing and feature extraction

RXTE gives three kinds of input from the Proportional Counter Array (PCA; [Jahoda et al. 2006](#)): Standard-2 spectra, PCA Standard-1 light curves and, for one test, PCA event files (HEXTE spectra are a fourth, minor input). Spectra are read and background-subtracted in Python (`spectra.py`); HEASoft is used only for an independent check. Figure 5 shows the spectral chain and Figure 7 the timing chain.



**Figure 5:** RXTE/PCA spectral pipeline. Top row: processing of every observation; the quality rules decide whether the next candidate of the same time group is tried. Middle: the representations built from the accepted spectrum, each with the round that added it. Bottom: HEXTE colours (v4b3).

### 4.1 Standard-2 Standard Products: what is read

For each ObsID three files from `stdprod/` are used (Table 4). They were checked on two observations (`05_two_source_check.py`, `reports/step2_fits_check.md`):

- `xp<0>_s2.pha` is the total (source + background) spectrum: `HUCLAS2/3 = TOTAL/COUNT`; column `COUNTS` in counts with an explicit `STAT_ERR` (`POISSERR = F`); `SYS_ERR = 0`; `BACKSCAL = AREASCAL = 1`; 129 channels. The PCUs summed are listed in the `ROWID<n>` keywords.
- `xp<0>_b2.pha` is the `pcabackest` model background for the same PCUs and good time intervals (GTIs). Its error is the statistical error of the model counts; the systematic error of the background model is not used anywhere.
- `xp<0>_rsp` is the combined `RMF×ARF` for that PCU combination, with `EBOUNDS`.
- The Standard Products are *not* dead-time corrected.

`spectra.read_pha` returns the counts, errors, exposure, scaling keywords, PCU list, pointing (`RA_OBJ`, `DEC_OBJ`), times and the `STDGTI` table. `spectra.read_rsp` expands the compressed response rows (`F_CHAN`, `N_CHAN`, `MATRIX`) into a dense matrix  $R(E, c)$ .

### 4.2 Net count-rate spectrum

For channel  $c$  (`spectra.net_spectrum`), with source counts  $C_c^s \pm \sigma_c^s$  and background counts  $C_c^b \pm \sigma_c^b$ :

$$t_s = \text{EXPOSURE}_s \text{AREASCAL}_s, \quad t_b = \text{EXPOSURE}_b \text{AREASCAL}_b, \quad \kappa = \text{BACKSCAL}_s / \text{BACKSCAL}_b, \\ r_c^s = C_c^s / t_s, \quad r_c^b = \kappa C_c^b / t_b, \quad r_c^{\text{net}} = D(r_c^s - r_c^b), \quad e_c^{\text{net}} = D\sqrt{(\sigma_c^s / t_s)^2 + (\kappa \sigma_c^b / t_b)^2}, \quad (1)$$

in counts per second summed over the PCUs.  $D = 1/(1 - \text{DTF})$  is the dead-time correction (Section 4.3). This is the OGIP convention, the same quantity XSPEC uses after `data`, `backgrnd` and `response`. Negative net rates (faint sources at high energy) are kept as they are.

### 4.3 Dead time

- **v1.** Only the good-xenon term was available:  $\text{DTF} = 10^{-5} \text{ s} \times \sum_c C_c^s / (t_s N_{\text{PCU}})$ . This is a lower bound.
- **v2 onwards.** The RXTE GOF recipe is applied to the Standard-1 data. `spectra.std1_rates` reads every `FS46_*` file of the ObsID and averages the counts over the 0.125 s bins whose centres fall inside the spectrum's GTIs. This gives the good-xenon rate  $X_{e_p}$  of each PCU and the array rates  $V_p$  (propane),  $R_{em}$  (remaining) and  $VLE$  (very large events). With  $N_{\text{on}}$  the number of PCUs with  $X_{e_p} > 1 \text{ c s}^{-1}$  (`spectra.std1_dtf`):

$$\begin{aligned} \text{DTF}_p &= 10^{-5} \left( X_{e_p} + \frac{V_p + R_{em}}{N_{\text{on}}} \right) + 6 \times 10^{-5} \frac{VLE}{N_{\text{on}}}, \\ \text{DTF} &= \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \text{DTF}_p, \quad D = \frac{1}{1 - \text{DTF}}, \end{aligned} \quad (2)$$

where  $\mathcal{P}$  is the set of PCUs in the spectrum. A variant with  $1.5 \times 10^{-4} \text{ s}$  per VLE event is stored as `dtf_std1_vle150us` for sensitivity only.

- **Effect.** In the v2 sample, Standard-1 covers a median 99% of the spectrum GTIs, and the DTF has median 1.38% and maximum 8.8%.  $D$  scales the whole spectrum, so it changes intensities but not colours or shapes.

### 4.4 Common energy grid and rebinning

The channel-to-energy relation drifts with the gain, so every spectrum is moved onto one fixed grid. The grid comes from the response of a reference observation (`REFERENCE_OBS = 91702-01-66-05`). Its edges are the union of that response's `EBOUNDS` edges between the edge closest to 5 keV and the edge closest to 25 keV: 45 bins from 4.908 to 25.008 keV, 0.41–0.86 keV wide. (In v3b the lower limit is the edge closest to 3 keV: 50 bins from 2.883 keV.) For an observation with its own channel boundaries  $[E_k^{\text{lo}}, E_k^{\text{hi}}]$ , the overlap matrix (`spectra.overlap_matrix`)

$$W_{jk} = \frac{\max(0, \min(E_k^{\text{hi}}, e_{j+1}) - \max(E_k^{\text{lo}}, e_j))}{E_k^{\text{hi}} - E_k^{\text{lo}}}$$

is the fraction of channel  $k$  in grid bin  $j$ , assuming counts are spread uniformly within a channel. The rebinned rate and error per keV per PCU are

$$x_{ij} = \frac{\sum_k W_{jk} r_{ik}^{\text{net}}}{N_{\text{PCU},i} \Delta e_j}, \quad \sigma_{ij} = \frac{\sqrt{\sum_k W_{jk}^2 (e_{ik}^{\text{net}})^2}}{N_{\text{PCU},i} \Delta e_j}, \quad \Delta e_j = e_{j+1} - e_j. \quad (3)$$

The total and background spectra are rebinned in the same way (`srates`, `brates`). Within gain epoch 5 the channel shift relative to the reference is at most 0.85 channel (median 0.17); it is stored as `gain_shift_channels`.

**Summary quantities.** For each observation `04_fetch_process.py` computes

$$F_i = \sum_{j=0}^{44} x_{ij} \Delta e_j, \quad \sigma_{F,i} = \left( \sum_j (\sigma_{ij} \Delta e_j)^2 \right)^{1/2}, \quad \text{S/N}_i = F_i / \sigma_{F,i}, \quad (4)$$

as well as the 5–25 keV source and background rates, the background fraction, a 10–25/5–10 keV hardness and the number of non-positive bins. These quantities, the header keywords and the dead-time terms all go into `observations.csv`. The quality rules of Table 9 are then applied.

## 4.5 Response check and the folded power law

A fixed photon spectrum  $N(E) = E^{-2}$  is folded through each observation's response (`spectra.powerlaw_folded`) and rebinned like the data:

$$P_{ic} = \sum_{\ell} R_i(E_{\ell}, c) \int_{E_{\ell}^{\text{lo}}}^{E_{\ell}^{\text{hi}}} E^{-2} dE, \quad p_{ij} = \frac{\sum_c W_{jc} P_{ic}}{N_{\text{PCU},i} \Delta e_j} \quad (\text{key p12}). \quad (5)$$

The spread of  $p_{ij}$  across observations measures how much the response alone changes a count spectrum: median 3.7%, up to about 12%, depending on the PCU combination (`06_dataset_checks.py`). From v4b2 on,  $p_{ij}$  also defines the response-normalised representations  $R$  and  $H_R$ , which can be compared across gain epochs.

## 4.6 Independent check with XSPEC

`10_xspec_crosscheck.py` runs HEASoft 6.37.1 XSPEC in WSL on 12 randomly chosen v2 observations (seed 42), with `data`, `backgrnd`, `response`, `ignore **-5.0 25.0-**` and `tclout rate`. It compares the result with Equation (1) ( $D = 1$ ) summed over the same channels. Rates agree to  $3 \times 10^{-10}$  and errors to  $4 \times 10^{-10}$  (relative; `results/v2/xfspec_crosscheck.csv`). The check covers the background subtraction only, not the dead time or the rebinning.

## 4.7 Spectral representations

Every representation is computed from one observation's rebinned spectrum alone. Standardisation across observations happens later, inside the model (Section 7). Code: `07_train_eval.py`, `v3lib.v2_representations`, `12_v3b_extended_band.py`, `reps_R` in `22_v4b2_analysis.py`.

- **A** (intensity kept):  $A_{ij} = \text{asinh}(x_{ij}/a)$  with  $a = 0.1 \text{ c s}^{-1} \text{ keV}^{-1} \text{ PCU}^{-1}$  (`ASINH_SCALE`). The transform is log-like for bright bins and still defined for negative bins.
- **B** (shape):  $B_{ij} = x_{ij}/F_i$  ( $\text{keV}^{-1}$ ), so that  $\sum_j B_{ij} \Delta e_j = 1$ .
- **H** (two colours): band sums  $C_{i,m} = \sum_{j \in b_m} x_{ij} \Delta e_j$  over the bands of Table 11, and

$$\mathbf{h}_i = (h_{i1}, h_{i2}) = \left( \frac{C_{i,2}}{C_{i,1}}, \frac{C_{i,4}}{C_{i,3}} \right). \quad (6)$$

The colours are linear ratios. The plan specified  $\log_{10}$ , but two BH spectra had a slightly negative 16–25 keV sum, so the linear form was adopted before any model was trained (decision log). The denominators are positive for every observation, which the code asserts.

- **HI, H+B, HI+A**:  $(h_1, h_2, \log_{10} F)$ ;  $(\mathbf{h}, B_{i0}, \dots, B_{i44})$  (47 columns);  $(\text{HI}, A_{i0}, \dots, A_{i44})$  (48 columns).
- **B3, H3** (v3b, 50-bin grid):  $B$  normalised over 2.883–25.008 keV, and three colours  $(C_0/C_1, C_2/C_1, C_4/C_3)$  with  $C_0 = 2.883\text{--}4.908 \text{ keV}$ .
- **R, H<sub>R</sub>** (v4b2 onwards):  $R_{ij} = x_{ij}/p_{ij}$ , and colours of the data divided by the colours of the folded power law,  $h_1^R = (C_2/P_2)/(C_1/P_1)$  and  $h_2^R = (C_4/P_4)/(C_3/P_3)$ , with  $P_m = \sum_{j \in b_m} p_{ij} \Delta e_j$ .

## 4.8 HEXTE colours (v4b3)

`19_v4b3_hexte.py` downloads the cluster-B products `xh<0>_s1.pha` and `xh<0>_b1.pha`, plus the RMF and ARF named in their headers. Only cluster B is used, and only before 2009-12-14 16:10 UT (MJD 55179.6736), when it stopped rocking between source and background; 60 of the 456 v2 observations therefore have no HEXTE data. `DEADAPP = T` is required. The net spectrum follows

**Table 11:** Colour bands: grid bins whose geometric centre  $\sqrt{e_j e_{j+1}}$  lies in the nominal interval (`v3lib.band`).

| Band        | Nominal interval | Bins $j$                  | $n$ | Actual range [keV] |
|-------------|------------------|---------------------------|-----|--------------------|
| $b_1$       | [5, 7)           | 0–4                       | 5   | 4.908–6.948        |
| $b_2$       | [7, 10)          | 5–11                      | 7   | 6.948–9.827        |
| $b_3$       | [10, 16)         | 12–26                     | 15  | 9.827–16.082       |
| $b_4$       | [16, 25.1)       | 27–44                     | 18  | 16.082–25.008      |
| $b_0$ (v3b) | [2.8, 5)         | 5 bins of the 50-bin grid | 5   | 2.883–4.908        |

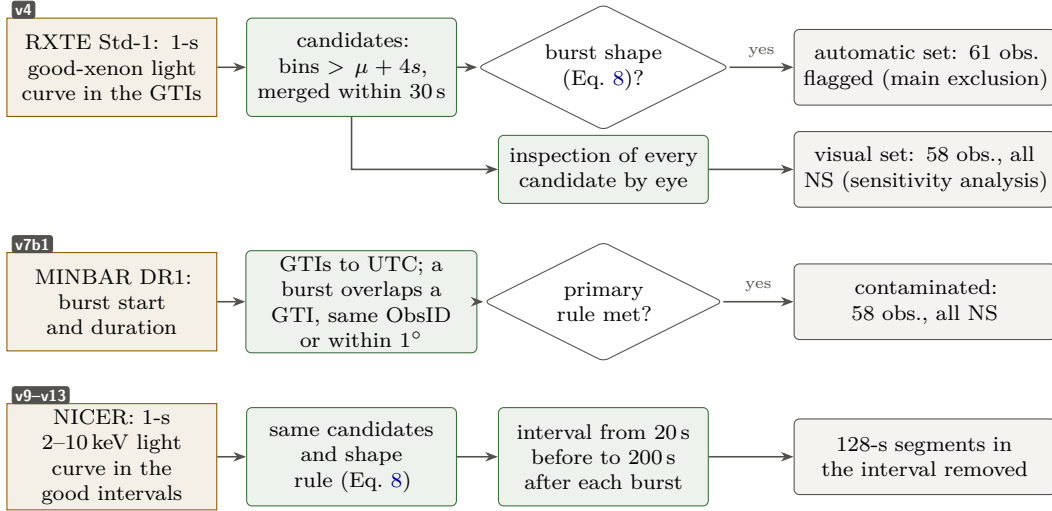
Equation (1) with  $D = 1$ , and band rates are computed with the overlap matrix on the RMF EBOUNDS. With  $F_{\text{PCA}}(16\text{--}25) = C_{i,4}$ :

$$X_1 = \frac{F_X(25\text{--}40)}{F_{\text{PCA}}(16\text{--}25)}, \quad X_2 = \frac{F_X(40\text{--}60)}{F_X(25\text{--}40)}. \quad (7)$$

A colour is set to missing when its denominator has  $S/N < 3$  (3 observations for  $X_1$ , 144 for  $X_2$ ). Inside each training fold, missing colours are replaced by the training median, and a 0/1 indicator column is added for each colour (Section 6.3). Output: `data/v4b3/processed/hexte_features.csv`.

#### 4.9 Type-I burst screening

Only neutron stars produce type-I bursts, so an observation containing one carries a label shortcut. Three procedures are used (Figure 6).



**Figure 6:** Burst screening: the Standard-1 detector (v4), the MINBAR match (v7b1) and the NICER rule (v9–v13), which removes segments rather than observations. Counts refer to the 456 v2 observations.

**Standard-1 detector (v4, `17_v4_burst_detect.py`).** The 1 s light curve  $r(t)$  is the sum of good-xenon counts of the spectrum’s PCUs over complete 1 s bins inside the GTIs. With  $\mu$  and  $s$  the mean and standard deviation of all bins, bins with  $r > \mu + 4s$  are candidates, and candidates closer than 30 s are merged. For an event with peak  $r_{\text{pk}}$  and persistent level  $r_0$  (median of the preceding 60 s),  $A = r_{\text{pk}} - r_0$  and  $L_q = r_0 + qA$ ,

$$\text{burst} \iff r_{\text{pk}} \geq 1.5 r_0 \wedge t_{\text{rise}} \leq 10 \text{ s} \wedge 3 \leq t_{1/2} \leq 300 \text{ s} \wedge n_{25} \geq 3, \quad (8)$$

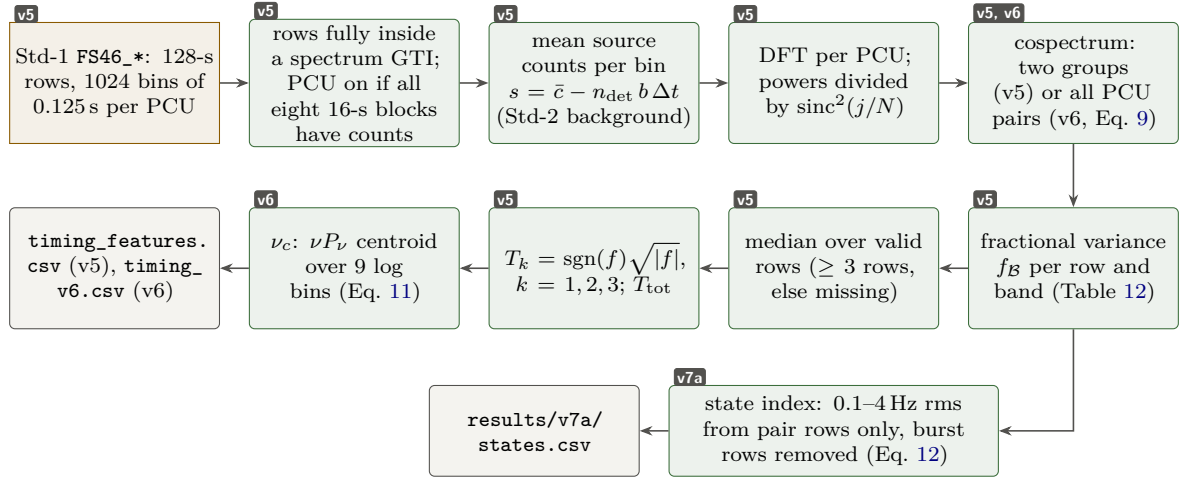
where  $t_{\text{rise}}$  is measured from the last time below  $L_{0.25}$  to the peak,  $t_{1/2}$  from the peak to the first time below  $L_{0.5}$ , and  $n_{25}$  counts the contiguous bins above  $L_{0.25}$  (`V4_BURST` in `config.py`). The burst interval runs from the last time before the peak below  $L_{0.25}$  to the first time after it below  $L_{0.1}$ . Every candidate was also inspected by eye (`results/v4/burst_visual.csv`).

**MINBAR match (v7b1, 31\_v7b1\_minbar.py).** The spectrum GTIs (mission time, TT) are converted to MJD(TT) and then to UTC with `astropy (v7lib.gti_utc)`. A MINBAR burst occupies  $[T, T + \text{dur}]$ , with 300 s when the duration is missing. Under the primary rule, an observation is contaminated if a burst interval overlaps a GTI and the burst either has the same ObsID or comes from within  $1^\circ$  (the PCA field of view) of the source. The literal planned rule (any burst from anywhere) is a sensitivity set. `32_v7b1_replace.py` builds a further sensitivity sample by replacing each contaminated observation with the next MINBAR-clean candidate of the same time group.

**How the flags are used.** In the RXTE analyses, flagged observations stay in the main sample and are removed only in sensitivity analyses (`18_v4_burst_exclusion.py`, `33_v7b1_exclusion.py`). The state index (below) is different: it drops every Standard-1 row that overlaps a burst interval, starting 20 s before the burst. For NICER, the segments near a detected burst are removed (Section 5.1).

#### 4.10 Low-frequency timing from Standard-1 (v5–v10)

Figure 7 shows the timing chain. It is shared, with different constants, by RXTE event mode and NICER.



**Figure 7:** Standard-1 timing chain (`v5lib.py`, `v6lib.py`, `v7lib.py`; scripts 26, 29, 34, 42, 53). No new download is needed: the Standard-1 files were already fetched for the dead time.

**Rows, detectors and background.** A Standard-1 row holds  $N = 1024$  bins of  $\Delta t = 0.125$  s (128 s) per PCU. Only rows lying completely inside a spectrum GTI are used, with a 1 ms tolerance (`v5lib.std1_rows`). A PCU is *on* in a row if each of the eight 128-bin blocks has at least one count (`v5lib.pcus_on`). The background rate per PCU is taken from the Standard-2 background spectrum,  $b = \sum_c C_c^b / (t_b N_{\text{PCU},b})$ , and assumed equal for all PCUs. For counts  $c_n$  summed over  $n_{\text{det}}$  detectors, the mean source counts per bin is  $s = \bar{c} - n_{\text{det}} b \Delta t$ ; a row with any  $s \leq 0$  is invalid.

**Fourier conventions.**  $X_j = \sum_n c_n e^{-2\pi i j n / N}$  for  $j = 0, \dots, N/2$  (`numpy.fft.rfft`), with  $\nu_j = j / (N \Delta t)$ , i.e.  $j / 128$  Hz. By Parseval's relation,  $\frac{2}{N^2} \sum_{j \in \mathcal{B}} |X_j|^2$  is the variance contributed by band  $\mathcal{B}$ . Binning multiplies the Fourier amplitude by  $\text{sinc}(j/N) = \sin(\pi j/N) / (\pi j/N)$ , so every power is divided by  $\text{sinc}^2(j/N)$ .

**Cospectra instead of power spectra.** Counting noise in different detectors is uncorrelated, so the *cross* spectrum between detectors has zero expected value without source variability. The Poisson and dead-time noise level therefore never has to be modelled.

- **v5 (two groups, v5lib.row\_features).** The PCUs that are on are sorted; those in even positions form group A and those in odd positions group B (at least 2 PCUs). With summed counts  $x, y$ , transforms  $X, Y$  and source means  $s_x, s_y$ :  $f_B = \frac{2}{N^2} \sum_{j \in B} \Re(X_j Y_j^*) / \text{sinc}^2(j/N) / (s_x s_y)$ .
- **v6 (all pairs, v6lib.allpairs\_row).** Every pair of the  $n$  PCUs that are on is used, through the closed form

$$f_B = \frac{\frac{2}{N^2} \sum_{j \in B} \frac{|\sum_p X_{p,j}|^2 - \sum_p |X_{p,j}|^2}{2 \text{sinc}^2(j/N)}}{\frac{(\sum_p s_p)^2 - \sum_p s_p^2}{2}}. \quad (9)$$

Rows with one PCU on use the auto-power with a Poisson level,  $(|X_j|^2 - N\bar{c}) / \text{sinc}^2(j/N)$  divided by  $s^2$ , but only below  $500 \text{ cs}^{-1}$ ; brighter single-PCU rows are invalid. In synthetic tests the noise variance falls by 31–41% for 3–5 PCUs, without bias.

**From rows to features.** For each observation, the median over valid rows is taken in each band; with fewer than 3 valid rows the features are missing. The features are signed square roots (fractional rms, slightly negative when the signal is below the noise):

$$T_k = \text{sgn}(\text{median}_r f_{k,r}) \sqrt{|\text{median}_r f_{k,r}|} \quad (k = 1, 2, 3),$$

$$T_{\text{tot}} = \text{sgn}(F) \sqrt{|F|}, \quad F = \sum_k \text{median}_r f_{k,r}. \quad (10)$$

The bands are listed in Table 12. As a check, the v5 cospectral  $T_2$  agrees with the conventional auto-power estimate at Spearman 0.970 (276 observations with 20–500 source  $\text{cs}^{-1} \text{ PCU}^{-1}$ ).

**Characteristic frequency  $\nu_c$  (v6, v6lib.nu\_centroid).** Nine logarithmic Fourier-index bins  $[1, 1], [2, 3], [4, 7], \dots, [256, 511]$  are formed.  $P_b$  is the median over rows of the fractional variance in bin  $b$ ; the bin centre is  $\nu_b = \sqrt{j_{\text{lo}} j_{\text{hi}}} / 128 \text{ Hz}$  and its width in  $\ln \nu$  is  $\Delta_b = \ln((j_{\text{hi}} + 0.5) / (j_{\text{lo}} - 0.5))$ . Then

$$\ln \nu_c = \frac{\sum_b w_b \ln \nu_b}{\sum_b w_b}, \quad w_b = \frac{\max(P_b, 0)}{\Delta_b}, \quad (11)$$

missing if  $T_{\text{tot}}^* < 0.05$  or  $\sum_b w_b \leq 0$ .  $\nu_c$  is the frequency around which most variability power per logarithmic frequency interval lies: a model-free stand-in for a break frequency. (NICER uses 13 octave bins and does not divide by  $\Delta_b$ ; Section 5.1.)

**State index (v7a, v7lib.state\_rms).** The 0.1–4 Hz fractional rms ( $j = 13$ –511) is computed with Equation (9) from rows with at least two PCUs only, after removing every row that overlaps a burst interval  $[t_{\text{start}} - 20 \text{ s}, t_{\text{end}}]$  (MINBAR field-of-view rule and the v4 detector). With  $\rho = \text{sgn}(\text{median}_r f) \sqrt{|\text{median}_r f|}$ ,

$$\text{state} = \begin{cases} \text{hard-like} & \rho > 0.10 \\ \text{soft-like} & \rho < 0.075 \\ \text{intermediate} & \text{otherwise} \end{cases} \quad (\text{unknown with fewer than 3 rows}). \quad (12)$$

The thresholds are those of [Remillard and McClintock \(2006, Table 2\)](#), which are defined for 0.1–10 Hz. The 4 Hz Nyquist limit of Standard-1 makes  $\rho$  slightly low, so “hard-like” is conservative. A sanity gate must pass before the states are used: at least two thirds of the XTE J1118+480 observations must be hard-like, and two thirds of the Z-source observations (Cyg X-2, GX 17+2) soft-like.

#### 4.11 High-frequency timing from event mode (v8c)

`43_v8c_events.py` first surveys the `SE*.evt.gz` files in each `pca/` directory, reading only their headers with an HTTP range request. A file qualifies if DATAMODE matches `^E_\d+us_\d+M_0_\d+s$,` TDDSE2 contains all six anode chains, TIMEDEL  $\leq 2^{-12} \text{ s}$  and a PCUID column exists (`v8lib.mode_`

**Table 12:** Timing bands (Fourier index  $j$ ;  $\nu_j = j/128$  Hz for Standard-1 and NICER,  $j/16$  Hz for event mode).

| Band                | $j$         | Frequency [Hz] | Data         | Setting                  |
|---------------------|-------------|----------------|--------------|--------------------------|
| $T_1$               | 2–12        | 0.016–0.094    | Std-1, NICER | V5_BANDS_J, V9_T_BANDS_J |
| $T_2$               | 13–128      | 0.10–1.0       | Std-1, NICER |                          |
| $T_3$               | 129–511     | 1.0–4.0        | Std-1, NICER |                          |
| state (v7a)         | 13–511      | 0.10–4.0       | Std-1        | V7_RMS_J                 |
| state (v9+)         | 13–1280     | 0.10–10        | NICER        | V9_STATE_J               |
| white-noise monitor | 12288–16383 | 96–128         | NICER        | V9_WHITE_J               |
| LC                  | 16–63       | 1–4            | event mode   | V8_HF_BANDS_J            |
| HF1                 | 64–1023     | 4–64           | event mode   |                          |
| HF2                 | 1024–8191   | 64–512         | event mode   |                          |
| HF3                 | 8192–16383  | 512–1024       | event mode   |                          |
| NULL                | 24576–32767 | 1536–2048      | event mode   |                          |

qualifies). If several modes qualify, the one with the largest total size is downloaded (2.4 GB in total). Events are binned at  $2^{-12}$  s (Nyquist 2048 Hz) into 16 s segments ( $N = 65536$ ) that lie inside both the spectrum GTIs and the event-file GTIs and do not overlap a burst. A PCU is on if each 1 s sub-interval has an event, and  $\geq 2$  PCUs are required.

Per segment  $r$ , the band numerator is  $\text{num}_{\mathcal{B},r} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} (\text{all-pairs cross power})$  and the denominator  $\text{den}_r = \sum_{p < q} s_p s_q$ . Single 16 s segments of faint sources gave unstable ratios, so the observation value is the ratio of means (`v8lib.hf_summary`):

$$\hat{R}_{\mathcal{B}} = \frac{\sum_r \text{num}_{\mathcal{B},r}}{\sum_r \text{den}_r}, \quad \widehat{\text{SE}} = \frac{1}{\sum_r \text{den}_r} \sqrt{\frac{n}{n-1} \sum_r (\text{num}_{\mathcal{B},r} - \hat{R}_{\mathcal{B}} \text{den}_r)^2}. \quad (13)$$

This replaced the planned mean of per-segment ratios after an implementation check, before any model. The features are  $\text{sgn}(\hat{R})\sqrt{|\hat{R}|}$  for LC, HF1, HF2 and HF3. Quality requires at least 10 segments and  $|\hat{R}_{\text{NULL}}/\widehat{\text{SE}}| \leq 5$  in the empty 1536–2048 Hz band. After results were seen, a correlated white component was found; a post hoc correction subtracts the null-band power scaled by bandwidth. Output: `data/v8c/processed/hf_features.csv`.

## 4.12 Outputs of the RXTE chain

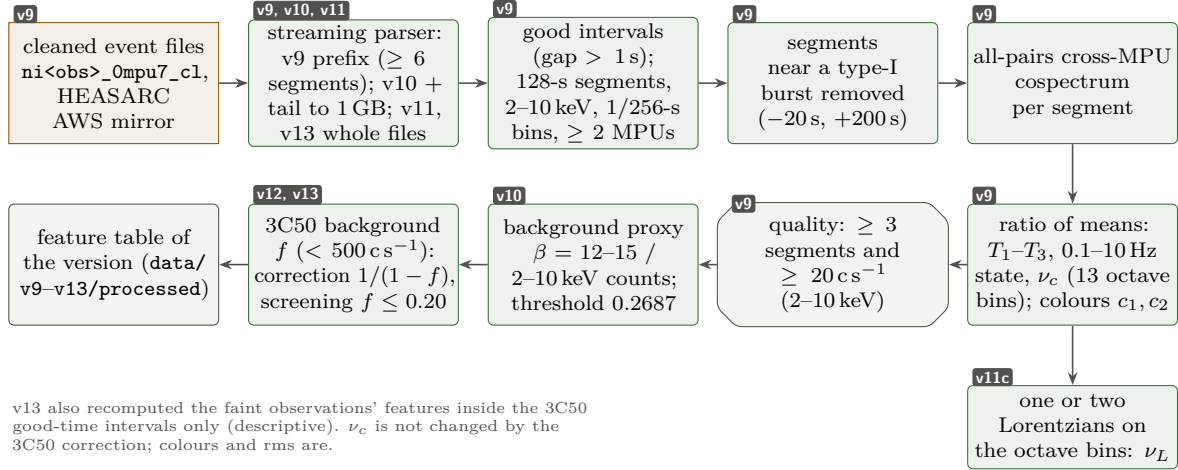
- `data/<v>/processed/features.npz`: rebinned spectra and errors of the accepted observations (Section 6).
- `data/<v>/observations.csv`: every tried candidate, with status, reason, header values, dead-time terms and summary rates.
- `data/<v>/processed/feature_definitions.csv`: grid edges and the definitions of A and B.
- Timing: `data/v5/processed/timing_features.csv`, `data/v6/processed/timing_v6.csv`, `data/v8b/processed/external_states_timing.csv`, `data/v10/processed/rxte_timing_table.csv`.
- Bursts and states: `results/v4/burst_flags.csv`, `results/v7b1/minbar_flags.csv`, `results/v7a/states.csv`.

## 5 NICER and MAXI processing

### 5.1 NICER/XTI: from event files to features (v9–v13)

The NICER chain (Figure 8) works on cleaned event files without HEASoft. It uses the same spectral idea as Standard-1, with the seven NICER measurement/power units (MPUs) as detectors.

HEASoft enters only for the 3C50 background (Section 5.2). Library code: `v9lib.py` (parser, segments, features), `v10lib.py` (tail and background proxy), `v11lib.py` (streaming, Lorentzians).



**Figure 8:** NICER/XTI feature pipeline: top row left to right, then bottom row right to left. The v9 steps were frozen before the observations of v11 were read, and the whole chain, including 3C50, before those of v13.

**Step 1: read the events.** `v9lib.PrefixParser` turns the compressed byte stream into event arrays (Section 2.3):

1. `zlib` (gzip mode) decompresses each new piece into a buffer;
2. once the buffer holds them, the primary and `EVENTS` headers are parsed from 2880-byte blocks;
3. `row_dtype` builds a big-endian numpy record type from `TFIELDS`, `TTYPEn` and `TFORMn`, checking that the row length equals `NAXIS1`;
4. complete rows are converted and only `TIME`, `PI` and `DET_ID` are kept;
5. the parse is *complete* when `NAXIS2` rows have been read.

Events are expected in time order; if they are not, they are sorted (and flagged).

**Step 2: good intervals and segments.** A good interval is a maximal stretch with no gap longer than 1 s between consecutive events of any energy (`v9lib.good_intervals`). Each interval  $[a, b]$  is cut into consecutive 128 s segments starting at  $a$ . Only 2–10 keV events (`PI` 200–999; `PI` is in units of 10 eV) are used for timing. The detector of an event is its MPU,  $m = \lfloor \text{DET\_ID}/10 \rfloor \in \{0, \dots, 6\}$ . Events are binned at 1/256 s ( $N = 32768$ , Nyquist 128 Hz). An MPU is on in a segment if each of the eight 16 s blocks contains a 2–10 keV event. A segment needs at least 2 MPUs on.

**Step 3: bursts.** The 1 s 2–10 keV light curve inside the good intervals (`v9lib.lc_1s`) is searched with the RXTE burst rule (Equation (8), `v9lib.burst_events`). Every segment that overlaps  $[t_{\text{start}} - 20 \text{ s}, t_{\text{end}} + 200 \text{ s}]$  of a detected burst is skipped; the number skipped is stored as `n_seg_burst_removed`.

**Step 4: per-segment statistics (`v9lib.segment_stats`).** For the  $n$  MPUs that are on, with mean counts per bin  $s_m$  (no background subtraction) and transforms  $X_{m,j}$ , the all-pairs cross power is  $\text{cross}_j = (|\sum_m X_{m,j}|^2 - \sum_m |X_{m,j}|^2) / (2 \text{sinc}^2(j/N))$  (Equation (9)). Each segment stores a numerator  $\text{num}_{B,r} = \frac{2}{N^2} \sum_{j \in B} \text{cross}_j$  for every band and for each of the 13 octave bins, and the denominator  $\text{den}_r = \sum_{m < m'} s_m s_{m'}$ .

### Step 5: observation features (`v9lib.obs_features`).

- **Timing.** Each band is combined over segments by the ratio of means with its delta-method error (Equation (13)), and converted to a signed square root. The bands are  $T_1$ ,  $T_2$ ,  $T_3$ , the state band 0.1–10 Hz and the 96–128 Hz white-noise monitor (Table 12).
- **Colours.** All events of all detectors inside the accepted segment windows are counted in PI 200–399 ( $A$ , 2–4 keV), 400–599 ( $B$ , 4–6 keV) and 600–999 ( $C$ , 6–10 keV);  $c_1 = B/A$  and  $c_2 = C/B$ . The colours therefore come from exactly the time windows used for timing.
- **Count rate.** `rate_2_10` is the mean over segments of the 2–10 keV counts of the MPUs that are on, divided by 128 s.
- $\nu_c$ . The 13 octave bins  $k = 0, \dots, 12$  cover  $j \in [2^k, 2^{k+1} - 1]$  (1/128–64 Hz). With  $\ell_k = \sum_r \text{num}_{k,r} / \sum_r \text{den}_r$ ,  $w_k = \max(\ell_k, 0)$  and  $\nu_k = 2^{k+0.5}/128$  Hz,  $\nu_c = \exp(\sum_k w_k \ln \nu_k / \sum_k w_k)$ . It is missing if  $\sqrt{\max(\sum_k \ell_k, 0)} < 0.05$ . Unlike the RXTE version (Equation (11)), the weights are not divided by the bin width; for octave bins this matters only for the lowest bin.
- **State.** Equation (12) is applied to the 0.1–10 Hz rms, the full band of Remillard and McClintock (2006).
- **Status.** `ok` needs at least 3 valid segments and `rate_2_10`  $\geq 20 \text{ cs}^{-1}$ ; otherwise `missing` or `excluded` (Table 9).

**Background proxy (`v10`, `v10lib.bkg_ratio`).**  $\beta = N(\text{PI } 1200\text{--}1499)/N(\text{PI } 200\text{--}999)$  over the whole file, the ratio of 12–15 keV to 2–10 keV events. Above 12 keV the source contributes little, so a high  $\beta$  indicates background. The threshold, frozen at 0.2687, is the 99th percentile of  $\beta$  among observations brighter than  $500 \text{ cs}^{-1}$ , where the background is negligible. It was used only for sensitivity analyses in `v10` and `v11`.

**Per-bin errors and Lorentzian fits (`v11c`, `v11lib`).** `lb_se` gives a delta-method error for each octave-bin variance  $\ell_k$ . `fit_lorentz` fits one or two zero-centred Lorentzians whose integrated variance in bin  $[\nu_k^{\text{lo}}, \nu_k^{\text{hi}}]$  (edges  $(j \mp 0.5)/128$  Hz) is

$$L_k(r^2, \Delta) = r^2 \frac{2}{\pi} \left[ \arctan \frac{\nu_k^{\text{hi}}}{\Delta} - \arctan \frac{\nu_k^{\text{lo}}}{\Delta} \right], \quad \Delta \in [1/128, 128] \text{ Hz}. \quad (14)$$

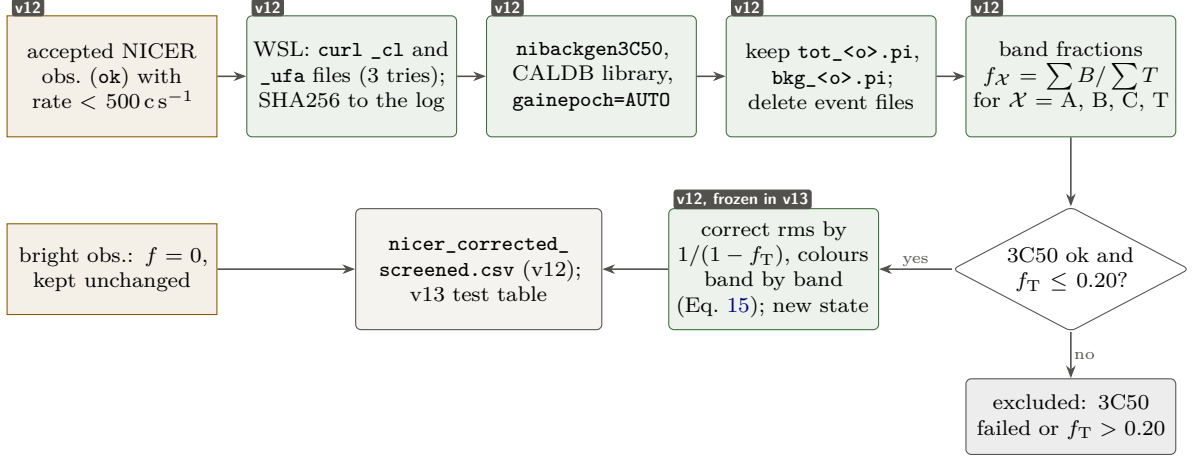
The fit is weighted least squares in log-parameters (`scipy.optimize.least_squares`), with four starting points per model; at least 6 usable bins are required. With  $\text{BIC} = \chi^2 + k \ln n$ , two components are chosen if  $\text{BIC}_1 - \text{BIC}_2 > 6$ . The width of the lowest-frequency component is  $\nu_L$ .

**Reading modes and storage.** `v9` parsed only a prefix of each file, stopping at 6 segments (Table 5). `v10` appended the tail up to 1 GB and re-parsed prefix and tail as one stream (`v10lib.read_full`); 366 of the 368 N1 observations remained `ok`. `v11` and `v13` streamed whole files into the frozen `obs_features` without storing them (`v11lib.obs_features_stream`), with 64 MB requests in windows of 6 concurrent requests consumed in order. The fetch scripts keep a checkpoint (`observations_partial.jsonl`), so an interrupted run resumes without re-reading finished observations.

## 5.2 NICER background: 3C50 correction and screening (`v12`, `v13`)

Figure 9 shows the background step. It is applied only to observations fainter than  $500 \text{ cs}^{-1}$  in 2–10 keV; for brighter ones the background is treated as negligible ( $f = 0$ ).

**Running 3C50.** The 3C50 model (Remillard et al., 2022) estimates the background spectrum of an observation from a library of blank-sky observations, selected by background-sensitive count rates measured in the observation itself (hence the unfiltered `_ufa` file). For each faint observation the Python driver calls, through `ws1 -d Ubuntu-22.04`, the shell script `v12_bkg3c50.sh` with the



**Figure 9:** 3C50 background step (58\_v12\_bkg.py, 62\_v13\_bkg.py, v12\_bkg3c50.sh; correction in corrected of 59\_v12\_analysis.py, copied unchanged into 63\_v13\_analysis.py).

ObsID, the cleaned and unfiltered URLs, an output folder and the CALDB folder. The script activates the HEASoft conda environment and points CALDB, CALDBCONFIG and CALDBALIAS at the unpacked library. It downloads the two files into /tmp/bkg3c50/<obsid>/xti/event\_c1/ and unzips them. It then runs

```
nibackgen3C50 rootdir=/tmp/bkg3c50 obsid=<obsid>
                bkgidxdir=CALDB bkglibdir=CALDB gainepoch=AUTO clobber=yes chatter=1
                totspec=tot_<obsid> bkgspec=bkg_<obsid>
```

copies the two spectra out and deletes the working folder. The driver reads the spectra (RATE, or COUNTS divided by EXPOSURE) and sums them in the PI bands A, B, C and T (200–999). It records the band rates  $T_\chi$  and  $B_\chi$ , the fractions  $f_\chi = B_\chi/T_\chi$  and the 3C50 exposure in data/v12/processed/bkg3c50.csv. Six workers run in parallel, and a checkpoint file makes the step resumable.

**Failures.** In v12, 362 of the 503 faint observations gave spectra; 141 failed, mostly because the tool rejects data processed with the oldest gain calibration. In v13, 109 of 148 succeeded. Failed observations are excluded from the corrected analyses.

**Correction and screening.** An observation is kept if it is bright, or if 3C50 succeeded with  $f_T \leq 0.20$ ; thresholds 0.10 and 0.30 are sensitivity variants. For kept faint observations,

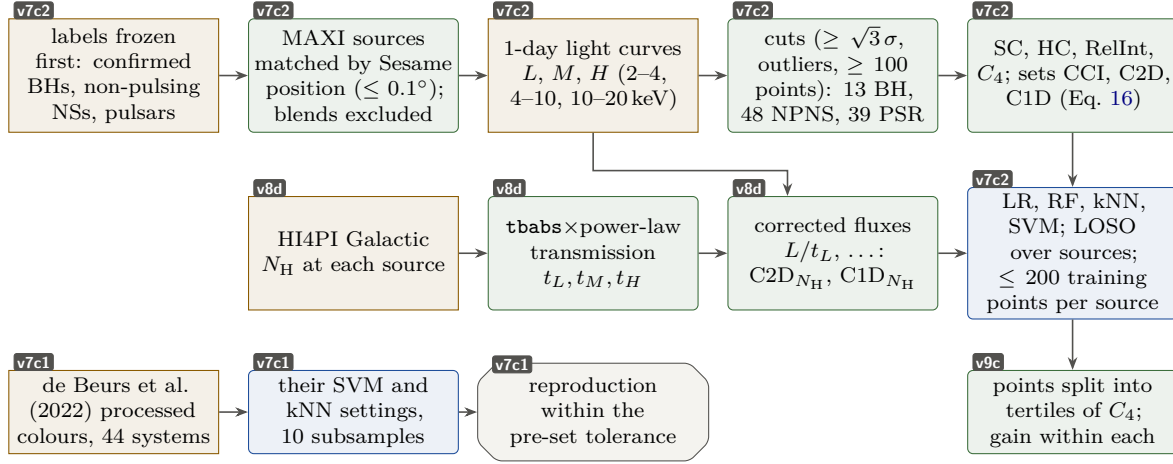
$$T'_k = \frac{T_k}{1 - f_T} \quad (k = 1, 2, 3, \text{state}), \quad c'_1 = \frac{B(1 - f_B)}{A(1 - f_A)}, \quad c'_2 = \frac{C(1 - f_C)}{B(1 - f_B)}, \quad (15)$$

and the state is re-classified with the corrected state rms. A fractional rms is normalised to the total mean count rate; renormalising it to the source mean  $S = (1 - f)(S + B)$  multiplies it by  $1/(1 - f)$ , assuming a constant background.  $\nu_c$  is not changed, because it depends only on the shape of the power spectrum. A check requires the 3C50 total 2–10 keV rate to be within 0.7–1.3 of the pipeline rate for at least 90% of observations.

**GTI-restricted variant (v13, descriptive).** For the faint v13 observations, the cleaned event files were kept (V13\_KEEP\_CL). Their features were recomputed using only events inside the GTIs of the 3C50 total spectrum (gti\_restricted in 63\_v13\_analysis.py), to check that the two time selections give the same picture.

### 5.3 MAXI/GSC: daily light curves (v7c, v8d, v9c)

MAXI points are days, not pointed observations. Figure 10 shows the chain.



**Figure 10:** MAXI/GSC pipeline: the strict pipeline (v7c2), the absorption correction (v8d), the colour strata (v9c) and the reproduction of [de Beurs et al. \(2022\)](#) (v7c1), which uses their processed colours instead of this pipeline.

### Source matching (38\_v7c2\_select.py).

1. The label lists are frozen first: 25 BHs ([Marcel et al., 2026](#)), the 115 MINBAR bursters and the pulsar lists. Sources with a pulse period in the Liu et al. HMXB or LMXB catalogues (CDS J/A+A/455/1165 and J/A+A/469/807) or in Patruno & Watts form the separate pulsar class. MINBAR bursters without a pulse period are the “non-pulsing” NSs (NPNS).
2. The MAXI source index is scraped from the 13 category pages [http://maxi.riken.jp/top/lc\\_<cat>.html](http://maxi.riken.jp/top/lc_<cat>.html). Each entry gives a name and a J-identifier, whose digits encode an approximate position (RA hhmm, Dec  $\pm$ dd.d).
3. Entries whose name contains “with” are blends of two sources and are removed. The remaining MAXI names are resolved with Sesame. A label source matches a MAXI source within  $0.1^\circ$  of the Sesame position, or within  $0.3^\circ$  of the J-name position if Sesame fails.
4. A MAXI source matched by label sources of different classes, or by several label sources that are not aliases of one object, is excluded as ambiguous (`data/v7c/c2_sources_ambiguous.csv`).

**Light curves and cuts.** For every matched source the 1-day light curve `<J>_g_lc_1day_all.dat` is downloaded (columns MJD, total flux and error, then  $L$ ,  $M$ ,  $H$  with errors for 2–4, 4–10 and 10–20 keV). The cuts of [de Beurs et al. \(2022\)](#) are then applied:

- every band must have flux/error  $\geq \sqrt{3}$  and a positive error;
- points with  $L + M + H \geq \text{mean} + 10 \text{sd}$  of the source are dropped;
- the source needs at least 100 remaining points.

Result: `data/v7c/maxi_points.csv` with 132057 points of 13 BH, 48 NPNS and 39 pulsar sources.

**Features (39\_v7c2\_analysis.py).** Per point,

$$\text{SC} = \frac{M - L}{M + L}, \quad \text{HC} = \frac{H - L}{H + L}, \quad C_4 = \frac{H - M}{H + M}, \quad \text{RelInt} = \frac{L + M + H}{Q_{99.99}(L + M + H)}, \quad (16)$$

where  $Q_{99.99}$  is the 99.99th percentile over that source’s points. The feature sets are CCI = (SC, HC, RelInt), C2D = (SC, HC) and C1D = ( $C_4$ ), the only colour without the 2–4 keV band.

### Absorption correction (v8d, 45\_v8d\_nh.py, v8d\_heasoft.sh).

1. HEASoft `nh` gives the HI4PI column at each source position (`disio=0.1`; the weighted average `avwnh` is used).

2. XSPEC computes photon fluxes of `tbabs*powerlaw` ( $\Gamma = 2$ ; `abund wilm`, `xsect vern`; a dummy response from 0.5 to 30 keV) in the three MAXI bands. The grid is  $N_{\text{H}} = 0$  plus 200 logarithmic steps from  $10^{19}$  to  $10^{23.5} \text{ cm}^{-2}$ .
3. The transmission is  $t_{\mathcal{X}}(N_{\text{H}}) = \Phi_{\mathcal{X}}(N_{\text{H}})/\Phi_{\mathcal{X}}(0)$ , interpolated linearly in  $\log N_{\text{H}}$ .
4. The corrected fluxes  $L/t_L$ ,  $M/t_M$ ,  $H/t_H$  give  $\text{SC}'$ ,  $\text{HC}'$ ,  $C'_4$  and  $\text{RelInt}'$  through Equation (16) (sets  $\text{C2D}_{N_{\text{H}}}$ ,  $\text{C1D}_{N_{\text{H}}}$ ,  $\text{CCI}_{N_{\text{H}}}$ ).

**Colour strata (`v9c`, `50_v9c_maxi_strata.py`).** The points are split into tertiles of  $C_4$ , and the models are compared within each tertile.

**Reproduction of de Beurs et al. (`v7c1`, `37_v7c1_repro.py`).** Their processed colours of 44 systems and their ten 20% subsamples are used with their three-class settings (BH, NPNS, pulsar): kNN with  $k = 24$  and SVM with  $C = 0.655$ ,  $\gamma = 0.585$ . The reproduction counts as successful if, for each algorithm, the number of correctly classified systems is within 2 of the published number (kNN 37, SVM 36) and the number of correct BHs within 1 (6 and 5) (`V7_REPRO_TOL_SOURCES`, `V7_REPRO_TOL_BH`). Otherwise the script stops.

## 6 Data model: from stored tables to design matrices

Every model in the project sees the same three objects: a design matrix, a label vector and a source vector. This section defines them, shows how they are assembled from the stored files, and lists the controls that keep information about a test source out of its own prediction.

### 6.1 Notation

An analysis uses  $n$  observations of  $S$  sources.

- $\mathbf{X} \in \mathbb{R}^{n \times p}$ , the **design matrix**. Row  $i$  is one observation: an RXTE ObsID, a NICER ObsID or one MAXI day. Column  $\ell$  is one feature.
- $\mathbf{y} \in \{0, 1\}^n$ , the labels,  $y_i = 1$  for a BH. Every observation inherits the label of its source.
- $\mathbf{g} \in \{1, \dots, S\}^n$ , the source (group) of each row;  $n_k$  is the number of rows of source  $k$ , and  $S_1$  and  $S_0$  are the numbers of BH and NS sources.
- $\mathbf{w}$ , the sample weights used in a fit (Section 7.2).
- $\hat{s}_i \in [0, 1]$ , the **out-of-fold (OOF) score** (“BH score”), always produced by a model fitted without any observation of source  $g_i$ . Scores are model outputs, not calibrated probabilities.
- **Source score and decisions:**

$$\bar{s}_k = \frac{1}{n_k} \sum_{i: g_i=k} \hat{s}_i, \quad \hat{Y}_k = \mathbf{1}[\bar{s}_k \geq t], \quad \hat{y}_i = \mathbf{1}[\hat{s}_i \geq t], \quad t = 0.5. \quad (17)$$

The only exception is the nested procedure of `v4a`, which selects a threshold inside the training sources (Section 7.7). Observations of one source are never multiplied together as independent evidence.

### 6.2 Assembling a design matrix

Algorithm 2 is the recipe that every analysis script follows. Two rules matter most:

- anything that uses more than one observation (standardisation, imputation, residualisation, thresholds, hyperparameters) is computed inside the training part of a fold;
- tables are joined by observation ID, never by row position, unless the code asserts that the orders are identical.

---

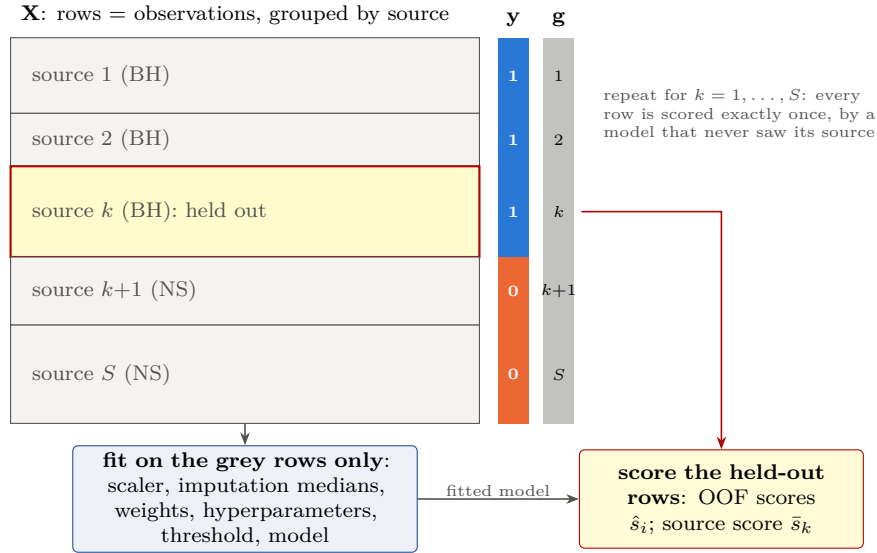
**Algorithm 2** Building  $(\mathbf{X}, \mathbf{y}, \mathbf{g})$  and the out-of-fold scores for one analysis.

---

**Require:** data/<v>/processed/features.npz; optional feature tables keyed by obs\_id (NICER: obsid)

- 1: load `rate`, `edges`, `F`, `pl2`, `y`, `source_id`, `obs_id` (one row per accepted observation)
- 2: compute the representation of each row (`A`, `B`, `H`, `HI`, `R`, `HR`; Section 4.7) ▷ uses that row only
- 3: to combine samples (e.g. v2 and v4b2), stack the rows and assert identical `edges`
- 4: join timing, HEXTE and state columns with `table.reindex(obs_id)`; non-finite values become NaN
- 5:  $\mathbf{X} \leftarrow$  column-stack of the chosen blocks;  $\mathbf{y} \leftarrow \mathbf{y}$ ;  $\mathbf{g} \leftarrow \mathbf{source\_id}$
- 6: `folds`  $\leftarrow$  `LeaveOneGroupOut().split(X, y, g)`; assert one test source per fold, disjoint sources and both classes in training
- 7: **for** each fold  $(\mathcal{T}, \mathcal{V})$  **do**
- 8:   fill missing values with medians of  $\mathbf{X}_{\mathcal{T}}$  and add indicators (Equation (18))
- 9:   fit the model on  $\mathcal{T}$  (scaler inside the model; weights from  $\mathbf{y}_{\mathcal{T}}, \mathbf{g}_{\mathcal{T}}$ ) and score  $\mathbf{X}_{\mathcal{V}}$
- 10: **end for**
- 11: write the OOF table: representation, model, fold, `source_id`, `obs_id`, true label, `BH_score`, predicted label

---



**Figure 11:** One leave-one-source-out fold. The held-out block holds all observations of one source; every learned quantity comes from the other sources.

### 6.3 Missing values

Spectral features are never missing: an observation that fails the quality rules is not in the sample. Timing features can be missing, for example with fewer than 3 valid rows. For a block  $T$  with training rows  $\mathcal{T}$  (`v5lib.fill_missing`):

$$m_\ell = \text{median}_{i \in \mathcal{T}, T_{i\ell} \text{ finite}} T_{i\ell} \quad (0 \text{ if none}), \quad \tilde{T}_{i\ell} = \begin{cases} T_{i\ell} & \text{finite} \\ m_\ell & \text{otherwise,} \end{cases} \quad (18)$$

$$\mu_i = \mathbf{1}[\exists \ell : T_{i\ell} \text{ not finite}].$$

$[\tilde{T}, \mu]$ , with one indicator column, replaces  $T$  in training and test rows. HEXTE colours use the same rule with one indicator per colour. The CCTLR classifier needs no imputation: its timing term is set to zero for a row with a missing value (Section 7.8). NICER features are never imputed, because observations that fail the NICER rules are excluded. For the frozen external tests (v6a, v8b), the medians come from the S1 training set and are applied unchanged.

## 6.4 Design matrices used by the analyses

Table 13 lists the main design matrices. The columns are the features defined in Sections 4 and 5, and  $m$  is a missing-value indicator.

**Table 13:** Main design matrices (shape  $n \times p$ ) and their columns.

| Round         | Matrix                                                 | Shape                                                       | Columns; source of the rows                                                      |
|---------------|--------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------|
| v1            | $\mathbf{X}_A, \mathbf{X}_B$                           | $120 \times 45$                                             | $A_{ij}$ or $B_{ij}$ , $j = 0..44$ ;<br><code>data/processed/features.npz</code> |
| v2            | $\mathbf{X}_A, \mathbf{X}_B$                           | $456 \times 45$                                             | as v1; <code>data/v2/processed/features.npz</code>                               |
|               | $\mathbf{X}_H, \mathbf{X}_{HI}$                        | $456 \times 2, 456 \times 3$                                | $h_1, h_2$ (+ $\log_{10} F$ )                                                    |
| v3a           | $\mathbf{X}_{H+B}, \mathbf{X}_{HI+A}$                  | $456 \times 47, 456 \times 48$                              | colours with the full shape or intensity spectrum                                |
| v3b           | $\mathbf{X}_{B3}, \mathbf{X}_{H3}$                     | $456 \times 50, 456 \times 3$                               | 50-bin grid from 2.883 keV;<br><code>features_3_25.npz</code>                    |
| v4b2          | $\mathbf{X}_H, \mathbf{X}_{H_R}, \mathbf{X}_R$         | $677 \times \{2, 2, 45\}$                                   | S1 + v4b2 rows; R uses p12                                                       |
| v4b1          | $\mathbf{X}_H \dots \mathbf{X}_A$                      | $7949 \times \{2, 3, 45, 45\}$                              | all epoch-5 pointings;<br><code>data/v4b1/processed/features.npz</code>          |
| v4b3          | $\mathbf{X}_{H+X}$                                     | $396 \times 6$                                              | $h_1, h_2, X_1, X_2$ + one indicator per HEXTE colour                            |
| v5            | $\mathbf{X}_{H+T}, \mathbf{X}_T$                       | $456 \times 6, 456 \times 4$                                | $h_1, h_2, T_1, T_2, T_3, m$ ; joined from<br><code>timing_features.csv</code>   |
| v6a           | $\mathbf{X}_{H_R}, \mathbf{X}_{H_R+T}$                 | train $456 \times \{2, 6\}$ , test<br>$341 \times \{2, 6\}$ | external set E; S1 medians for missing $T$                                       |
| v7c2,<br>v8d  | $\mathbf{X}_{CCI}, \mathbf{X}_{C2D}, \mathbf{X}_{C1D}$ | $93\,461 \times \{3, 2, 1\}$                                | one row per MAXI day (BH vs NPNS, 61 sources)                                    |
| v8b           | $\mathbf{X}_{H_R}, \mathbf{X}_{H_R+T}$                 | $379 \times \{2, 6\}$                                       | external set X, frozen S1 models; 99 hard-like rows                              |
| v8c           | $\mathbf{X}_{H+HF}$                                    | $148 \times 5$                                              | $h_1, h_2, HF_1, HF_2, HF_3$                                                     |
| v9a           | $\mathbf{X}_{H_N}, \mathbf{X}_{H_N+T_N}$               | $368 \times \{2, 5\}$                                       | $c_1, c_2$ (+ $T_1, T_2, T_3$ ); no missing values                               |
| v10a          | as v9a                                                 | $366 \times \{2, 5\}$                                       | full files                                                                       |
| v11           | as v9a                                                 | test $318 \times \{2, 5\}$                                  | new observations, frozen out-of-source models                                    |
| v12           | as v9a                                                 | $534 \times \{2, 5\}$                                       | corrected and screened N1+N2                                                     |
| v13           | as v9a                                                 | test $198 \times \{2, 5\}$                                  | corrected and screened N3; 74 hard-like rows                                     |
| $\nu_c$ tests | $(\mathbf{h}, \log_{10} \nu_c)$ per<br>hard-like obs.  | –                                                           | input of the permutation test (Section 8.4),<br>not a classifier                 |

## 6.5 Leakage controls in the code

1. **Folds by source.** `LeaveOneGroupOut` on `g`. The scripts assert that each test fold holds exactly one source, that training and test sources are disjoint and that both classes are in training; the folds are saved (`results/v2/fold_checks.csv`, `results/v2/folds_loso.csv`).
2. **The same folds across rounds.** The v2 folds are reused and asserted identical in v3a, v4a and v5, so paired comparisons use the same test sets.
3. **Training-only preprocessing.** `StandardScaler` is part of the model object (`make_pipeline` or `V4Model.fit`). Imputation medians and source-equal weights are computed from training rows only.
4. **Nested selection.** Anything that is selected (hyperparameters, representations, thresholds) is chosen by an inner LOSO over the training sources, never on the test source (Section 7.7).
5. **Label-free residualisation.** The permutation tests regress timing on colour with a kNN that never sees labels, leaving the source out (Section 8.4).
6. **Frozen models.** External and new-observation tests use models, medians and rules fixed before the test data were read. In same-source tests, each source is scored by the model fitted without it (Section 8.6).

7. **Joins.** Feature tables are joined by observation ID. Where arrays are concatenated by position, the code asserts that the ID orders agree.

## 7 Machine-learning methods

### 7.1 Problem formulation and software

Each model is a binary classifier on observations: it maps a feature vector  $\mathbf{x}_i$  to a BH score  $s(\mathbf{x}_i) \in [0, 1]$ . The question is about *sources*, so every score is produced out of fold (no model sees the source it scores), and observation scores are averaged per source (Equation (17)). All models are scikit-learn 1.8.0 estimators (Pedregosa et al., 2011) (`requirements-lock.txt`) with the settings in `config.py`. They are wrapped in the pipelines of `07_train_eval.py` and `v3lib.py`, in `V4Model` (`v4lib.py`, the eight-algorithm search) and in `C2Model` (`v7lib.py`, MAXI). Table 14 lists them.

**Table 14:** Models and their roles. P = model of a planned primary comparison; B = baseline; D/E = descriptive or exploratory.

| Model                               | Rounds               | Code                                  | Role                         |
|-------------------------------------|----------------------|---------------------------------------|------------------------------|
| Logistic regression (LR)            | v1–v13               | 07, v3lib.MODELS,<br>V4Model, C2Model | P, B                         |
| Random forest (RF)                  | v1–v13               | as LR                                 | P (v8d, v12b, v13a), D       |
| Dummy (class prior)                 | v1, v2               | 07_train_eval.py                      | B                            |
| ExtraTrees, HistGB,<br>QDA, MLP     | v4a                  | V4Model                               | E                            |
| RBF SVM                             | v4a, v7c2; v7c1      | V4Model, C2Model, 37                  | E, reproduction              |
| kNN classifier                      | v4a, v7c2, v8d; v7c1 | V4Model, C2Model, 37                  | E, reproduction              |
| Nested automatic<br>selection       | v4a                  | v4lib.nested_fold                     | P (v4)                       |
| CCTLR                               | v6c                  | v6lib.CCTLR                           | D                            |
| kNN regression<br>(residualisation) | v6d, v9b–v13b        | v6lib.source_residuals                | inside the permutation tests |
| Frozen $\nu^*$ rule                 | v11, v13             | 57, 63                                | D                            |

### 7.2 Shared components

**Standardisation.** Scale-sensitive models (LR, SVM, kNN, QDA, MLP, the colour part of CCTLR, the kNN regression) standardise each feature with the training rows  $\mathcal{T}$  of the current fit:

$$z_{i\ell} = \frac{x_{i\ell} - \mu_\ell}{\sigma_\ell}, \quad \mu_\ell = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} x_{i\ell}, \quad \sigma_\ell^2 = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} (x_{i\ell} - \mu_\ell)^2 \quad (\sigma_\ell = 0 \Rightarrow 1), \quad (19)$$

and apply the same  $(\mu_\ell, \sigma_\ell)$  to the test rows (`StandardScaler` inside the pipeline). Tree models use the raw features.

**Class-balanced weights.** The main sample has 7 BH and 24 NS sources. With  $n$  training rows,  $n_c$  of them in class  $c$ , `class_weight='balanced'` gives each row

$$w_i^{\text{bal}} = \frac{n}{2 n_{y_i}}, \quad \sum_{i: y_i=c} w_i^{\text{bal}} = \frac{n}{2}. \quad (20)$$

QDA takes no weights and uses equal priors instead. kNN corrects its output to equal priors (Equation (23)), and the MLP receives  $w^{\text{bal}}$  as sample weights.

**Source-equal weights (v4a, v4b1, v5 S3, v6 S3).** When sources contribute very different numbers of observations (6 to 1329 per source in v4b1), balanced weights let the most-observed sources dominate. With  $S_c$  training sources in class  $c$  and  $n_k$  rows of source  $k$  (`v4lib.source_equal_weights`):

$$w_i^{\text{src}} = \frac{n}{2 S_{y_i} n_{g_i}}, \quad \sum_{i:g_i=k} w_i^{\text{src}} = \frac{n}{2 S_c} \quad (k \in c), \quad \sum_{i:y_i=c} w_i^{\text{src}} = \frac{n}{2}. \quad (21)$$

Every source of a class then has the same total weight, and both classes sum to  $n/2$ , the scale of the balanced weights. `class_weight` is set to `None` when these weights are passed. The first implementation normalised each class to 1 instead of  $n/2$ , which acted as a much stronger LR penalty. It was fixed after the v4a results and the affected parts were re-run; no conclusion changed (decision log).

**Scores.** `predict_proba[:,1]` for LR, RF, ExtraTrees, HistGB and QDA; the prior-corrected vote fraction for kNN; the mean over seeds for the MLP; the logistic function of the decision value for SVM and CCTLR. Scores rank observations and are thresholded at 0.5; they are not calibrated probabilities.

### 7.3 Logistic regression

LR is the main model: H-LR (two colours) is the reference that every later model is compared against, and H+T-LR is the primary model of v5, v6a, v8a/b, v9a, v10a and v11a. On standardised inputs,  $s(\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{z} + b)$  with  $\sigma(u) = 1/(1 + e^{-u})$ , and

$$(\hat{\mathbf{w}}, \hat{b}) = \arg \min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i \in \mathcal{T}} \omega_i [-y_i \log s(\mathbf{x}_i) - (1 - y_i) \log(1 - s(\mathbf{x}_i))], \quad (22)$$

with  $C = 1$ , L-BFGS, at most 5000 iterations, an unpenalised intercept and  $\omega_i = w_i^{\text{bal}}$  (or  $w_i^{\text{src}}$ ). With a few hundred rows and 2–6 features the penalty is weak. LR assumes log-odds that are linear in the standardised features, so a timing feature has one slope everywhere in the colour plane. This is the motivation for CCTLR (Section 7.8).

### 7.4 Random forest

RF is the second reference model in every round, and the primary model of v8d (MAXI), v12b and v13a. It averages  $B = 500$  trees:  $s(\mathbf{x}) = \frac{1}{B} \sum_b \hat{p}_b(\mathbf{x})$ , where  $\hat{p}_b$  is the weighted BH fraction in the leaf of tree  $b$ . Each tree is grown on a bootstrap sample (row weights  $\omega_i$  times the bootstrap counts) with CART splits that maximise the weighted decrease of the Gini impurity  $G = 1 - \sum_c \pi_c^2$ . At each node  $m = \lfloor \sqrt{p} \rfloor$  features are candidates, so with the two colours of H every split uses one randomly chosen colour, and with  $p = 6$  two. Trees grow until nodes are pure or a child would have fewer than 2 rows (`min_samples_leaf=2`). Balanced class weights are computed once per fold, and `random_state=42`. RF can follow non-monotone class regions, which matters for MAXI and for timing–colour interactions. With fewer than 10 BH sources, however, a leaf can memorise one source.

### 7.5 Further algorithms of the v4a search

- **Dummy** (`strategy='prior'`): returns the training BH fraction and predicts the majority class. Under LOSO the majority depends on the held-out source. In v1 (4+4 sources), holding out a BH leaves an NS majority and vice versa, so the Dummy balanced accuracy is 0, not 0.5.
- **ExtraTrees**: like RF but without bootstrap, and with one random threshold per candidate feature at each node.

- **HistGradientBoosting**: an additive model on the log-odds scale; each of 200 trees is fitted to the weighted gradients and Hessians of the log loss, with features pre-binned into at most 255 quantile bins. Learning rate 0.1, 15 leaves, at least 20 rows per leaf, no L2 term and no early stopping.
- **RBF SVM**: SVC with balanced class weights and  $K(\mathbf{z}, \mathbf{z}') = e^{-\gamma \|\mathbf{z} - \mathbf{z}'\|^2}$ . `gamma='scale'` gives  $\gamma = 1/(p \text{Var } \mathbf{Z})$ ; the v4a grid multiplies it by  $\kappa \in \{0.3, 1, 3\}$ . The score is  $\sigma(f(\mathbf{z}))$  of the decision value  $f$ . Platt scaling is not used, because it runs an internal random cross-validation.
- **kNN classifier**: the BH fraction  $\hat{p}$  among the  $k$  nearest standardised training rows (Euclidean, uniform weights), corrected to equal priors with the training BH fraction  $\pi$ :

$$s(\mathbf{x}) = \frac{\hat{p}/\pi}{\hat{p}/\pi + (1 - \hat{p})/(1 - \pi)}. \quad (23)$$

- **QDA**: Gaussian classes with regularised covariances  $\hat{\Sigma}_c = (1 - r)S_c + rI$  ( $r = 0.1$ ) and equal priors.
- **MLP**: one hidden layer of 32 ReLU units, Adam,  $\alpha = 10^{-3}$ , balanced sample weights. Early stopping holds out a stratified 20% validation split and restores the best weights. Because a small MLP is unstable, the score is the mean over seeds  $\{0, \dots, 4\}$  (outer fits) or  $\{0, 1, 2\}$  (inner fits).

A planned 1D convolutional network on representation B was skipped, as declared in the plan, because `torch` was not installed. Principal component analysis was never used: “PCA” in the code always means the Proportional Counter Array.

## 7.6 Hyperparameters

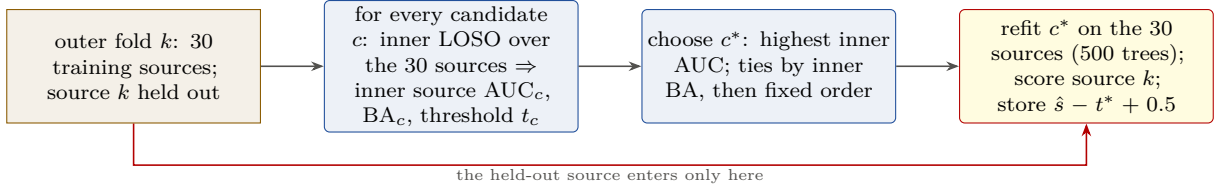
Except in the nested selection (Section 7.7) and the MAXI kNN and SVM, all hyperparameters are fixed in advance. Table 15 collects them.

**Table 15:** Hyperparameters (`config.py`: `LR_PARAMS`, `RF_PARAMS`, `V4_FIXED`, `V4_GRID`, `V7_C2_*`, `V7_DEBEURS_*`).

| Model                 | Fixed setting                                                                                                   | Grid (where tuned)                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| LR                    | $C = 1$ , L2, lbfgs, 5000 iterations, balanced                                                                  | v4a: $C \in \{0.01, 0.1, 1, 10\}$                                                                                              |
| RF                    | 500 trees, <code>max_features='sqrt'</code> , leaf $\geq 2$ , balanced, seed 42                                 | v4a: leaf $\in \{1, 2, 5, 10\}$ ; 200 trees inner, 500 outer                                                                   |
| ExtraTrees            | as RF                                                                                                           | v4a: leaf $\in \{1, 2, 5, 10\}$                                                                                                |
| HistGB                | learning rate 0.1, 200 iterations, 15 leaves, leaf $\geq 20$ , $\lambda = 0$ , no early stop, balanced, seed 42 | v4a: rate $\in \{0.05, 0.1\} \times$ leaves $\in \{7, 15\}$                                                                    |
| SVM (RBF)             | $C = 1$ , $\gamma = \text{'scale'}$ , balanced                                                                  | v4a: $C \in \{0.1, 1, 10\} \times \kappa \in \{0.3, 1, 3\}$ ; v7c2: $C \in \{0.1, 1, 10\} \times \gamma \in \{0.1, 0.585, 3\}$ |
| kNN                   | $k = 15$ , uniform, prior-corrected                                                                             | v4a: $\{5, 15, 31\}$ ; v7c2, v8d: $\{5, 15, 24, 35\}$                                                                          |
| QDA                   | $r = 0.1$ , priors (0.5, 0.5)                                                                                   | v4a: $r \in \{0.01, 0.1, 0.5\}$                                                                                                |
| MLP                   | (32, ), $\alpha = 10^{-3}$ , early stopping (20%), 500 epochs, seeds 0–4                                        | v4a: $\{(16, ), (32, 16)\} \times \alpha \in \{10^{-4}, 10^{-2}\}$                                                             |
| CCTLR                 | Student $t$ with 4 d.o.f., scale floor 0.01, ridge $10^{-9}$                                                    | none                                                                                                                           |
| kNN regression        | $K = 25$ , uniform, neighbours from other sources                                                               | none                                                                                                                           |
| de Beurs reproduction | kNN $k = 24$ on raw colours; SVM $C = 0.655298$ , $\gamma = 0.585144$ , Platt, one-vs-one, 3 classes            | none (taken from the paper)                                                                                                    |

## 7.7 Model selection by nested leave-one-source-out

When anything is selected (algorithm, representation, hyperparameter or threshold), the selection runs inside each outer training set, by an inner LOSO over its sources. The outer test source plays no part (Figure 12).



**Figure 12:** Nested LOSO (v4a). Every candidate is judged only on the outer training sources, so the cost of choosing the best of many is included in the outer estimate.

**v4a** (`v4lib.nested_fold`, `16_v4a_algorithms.py`). Four representations (H, HI, B, A) are crossed with eight algorithms and their grids (35 settings), giving 140 configurations. In part 1d, nesting runs per algorithm and representation; in part 1e, over all 140 configurations. The inner stage uses 200 trees and MLP seeds  $\{0, 1, 2\}$ , the outer refit 500 trees and seeds  $\{0, \dots, 4\}$ .

**Threshold choice** (`v4lib.choose_threshold`). With  $u_1 < \dots < u_q$  the distinct inner source scores, the candidates are  $u_1 - 10^{-9}$ , the midpoints  $(u_r + u_{r+1})/2$  and  $u_q + 10^{-9}$ . Among those maximising the inner source balanced accuracy, the one closest to 0.5 is taken:

$$t^* = \arg \min_{t \in \mathcal{C}^*} |t - 0.5|, \quad \mathcal{C}^* = \{t : \text{BA}_{\text{src}}(t) = \max_{t'} \text{BA}_{\text{src}}(t')\}. \quad (24)$$

The outer scores are stored shifted,  $\hat{s} - t^* + 0.5$ , so that the common 0.5 rule applies.

**MAXI** (`v7c2`, `v8d`; `v7lib.c2_outer`). kNN’s  $k$  is chosen by inner LOSO source AUC, and the SVM’s  $(C, \gamma)$  by inner `GroupKFold(5)` over sources (a pre-run change to save computing time). The first configuration with a strictly higher AUC wins. Each training source contributes at most 200 randomly chosen days (seed 42), and the test source is scored on all its days.

## 7.8 Colour-conditional timing likelihood ratio (CCTLR, v6c)

In LR a timing feature has one slope everywhere, although timing is expected to help only where the colours overlap. CCTLR adds to the H-LR logit a log-likelihood ratio of the timing features *given* the colours (`v6lib.CCTLR`):

$$\ell(\mathbf{x}) = \mathbf{w}^\top \mathbf{h} + b + \sum_{k=1}^3 \left[ \log t_4(T_k; \mu_{1k}(\mathbf{h}), \tau_{1k}) - \log t_4(T_k; \mu_{0k}(\mathbf{h}), \tau_{0k}) \right], \quad s = \sigma(\ell), \quad (25)$$

where  $t_4$  is the Student- $t$  density with 4 degrees of freedom (robust to outliers such as bursts or dips) and  $\mu_{ck}(\mathbf{h}) = \beta_{ck0} + \beta_{ck1}h_1 + \beta_{ck2}h_2$ . For each class and timing band,  $\beta_{ck}$  is a weighted least-squares fit on that class’s rows with finite timing. The weights are  $v_i = 1/(S_c n_{gi})$ , so every source counts equally, and a ridge of  $10^{-9}$  is added. The scale  $\tau_{ck}$  is the weighted residual standard deviation, floored at 0.01. If any timing value of a row is missing, the timing term is zero and the score equals the H-LR score. CCTLR has no tunable parameter.

## 7.9 kNN regression for colour residualisation

Inside every permutation test (Section 8.4), the part of a timing feature  $F$  (e.g.  $\log_{10} \nu_c$ ) that is predictable from colour is removed *without labels* (`v6lib.source_residuals`). With standardised

colours,

$$\hat{F}_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(\mathbf{h}_i; g_j \neq g_i)} F_j, \quad r_i = F_i - \hat{F}_i, \quad \bar{r}_k = \frac{1}{n_k} \sum_{i: g_i = k} r_i, \quad (26)$$

with  $K = 25$  (capped at the number of available rows). The neighbours of an observation come only from *other* sources, so a source cannot explain itself. Rows with missing  $F$  are skipped. If BHs cluster in one colour region, part of the class difference is absorbed into  $\hat{F}$ , which makes the test conservative.

### 7.10 Frozen characteristic-frequency rule (v11, v13)

A simple rule frozen before the new data were read: a hard-like observation is called BH if  $\nu_c < \nu^*$ , and a source is called BH if at least half of its new hard-like observations are. The threshold is the geometric mean of the class medians in the previous data,

$$\nu^* = \sqrt{\text{median}_{\text{BH}} \nu_c \cdot \text{median}_{\text{NS}} \nu_c}, \quad (27)$$

giving 1.3007 Hz from the v10 data (used in v11) and 1.3361 Hz from the corrected v12 data (used in v13). The values were written to `logs/v11/nu_star_frozen.txt` and `logs/v13/nu_star_frozen.txt` before the new observations were read.

### 7.11 Training designs

The same models are trained in five ways, depending on the question (Table 16).

**Table 16:** Training designs. “Out-of-source”: each source is scored by a model fitted on the reference data without that source.

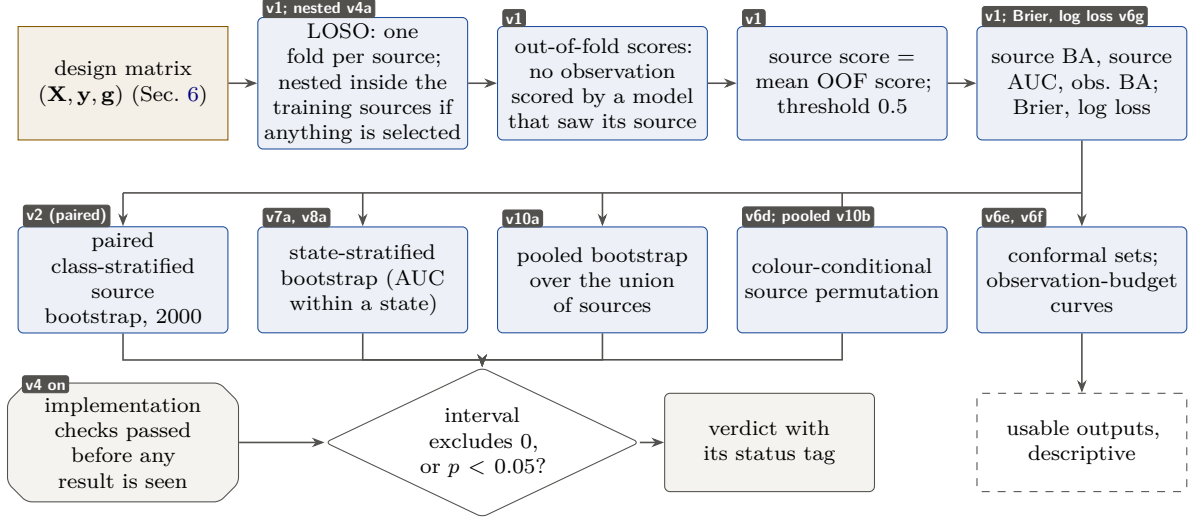
| Design               | What is fitted and what is scored                                                                                                               | Used in                                  |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| LOSO                 | one fit per source on all other sources; the held-out source is scored (Algorithm 2)                                                            | every development analysis (v1–v12)      |
| Nested LOSO          | as LOSO, plus inner LOSO selection within each training set (Section 7.7)                                                                       | v4a, v7c2, v8d                           |
| Frozen external      | one fit on S1; all external sources scored by the same model, with S1 imputation medians                                                        | v6a (set E), v8b (set X)                 |
| Frozen out-of-source | models fitted on the previous NICER data before the new data are read; each source’s new observations are scored by the fit without that source | v11 (reference v10), v13 (reference v12) |
| All-data model       | one fit on all reference data, for sources never used before                                                                                    | v11 (never-used sources)                 |

## 8 Evaluation and statistical inference

This section describes how a table of out-of-fold scores becomes a reported number with an interval, and how a difference is judged. Figure 13 shows the chain and Table 17 summarises the procedures. Every procedure works on *fixed* scores or features: nothing is refitted inside a resampling or a permutation.

### 8.1 Cross-validation and out-of-fold scores

Folds come from `LeaveOneGroupOut` on `g` (`v3lib.run_los`, `v4lib.los`). The OOF score of observation  $i$  is  $\hat{s}_i = s(\mathbf{x}_i; \hat{\theta}^{(-g_i)})$ , where  $\hat{\theta}^{(-k)}$  is fitted, with every learned step, on all rows of the other sources. For comparison only, v1 and v2 also ran a leaky split over observations (`StratifiedKFold(5, shuffle=True, random_state=42)`), in which a source appears in both training and test; it served only to show how optimistic such a split is.



**Figure 13:** Evaluation chain from the design matrix to a verdict, with the round that introduced each element. Bootstrap and permutation results are judged by the planned rule; conformal sets and budget curves are always descriptive.

**Table 17:** Inferential procedures. “Unit” is what is resampled or permuted.

| Procedure                         | Statistic                                          | Unit, size                              | Decision rule; code                                                                                                            |
|-----------------------------------|----------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Class-stratified source bootstrap | source and observation BA and AUC; Brier, log loss | sources within class; 2000, seed 42     | 95% percentile interval; a paired difference is detected if its interval excludes 0<br>( <code>v4lib.paired_bootstrap</code> ) |
| State-stratified bootstrap        | observation AUC within a state                     | sources of the whole sample; 2000       | as above ( <code>v8lib.strat_boot</code> )                                                                                     |
| Pooled bootstrap (v10a)           | mean of the instruments’ AUC gains                 | union of sources; 2000                  | as above ( <code>v10lib.pooled_boot</code> )                                                                                   |
| Colour-conditional permutation    | $D = \text{BH} - \text{NS}$ mean residual          | source labels; 20 000, seed 42          | two-sided $p < 0.05$ ; Holm across features ( <code>v6lib.perm_test</code> , <code>holm</code> )                               |
| Pooled permutation (v10b)         | mean of the instruments’ $D$                       | labels within membership strata; 20 000 | $p < 0.05$ ( <code>v10lib.pooled_perm</code> )                                                                                 |
| Mondrian conformal sets (v6e)     | per-class $p$ -values                              | calibration sources                     | $\text{set} = \{y : p_y > \varepsilon\}, \varepsilon \in \{0.1, 0.2\}$                                                         |
| Observation budget (v6f)          | source BA, AUC with $k$ observations per source    | observations within a source; 500 draws | median and 2.5–97.5% range                                                                                                     |

## 8.2 Metrics

- **Balanced accuracy:**  $\text{BA} = \frac{1}{2}(\text{TPR} + \text{TNR})$ . At *source level* it uses the decisions  $\hat{Y}_k$ ; with 7 BH and 24 NS sources, calling everything NS would already get 24 of 31 sources right, so BA, not accuracy, is reported. At *observation level* it uses the observation decisions; inside the bootstrap it is computed from per-source counts of correct observations.
- **ROC AUC** (Mann–Whitney form, ties counted as one half), for positive scores  $a_r$  and negative scores  $b_q$ :

$$\text{AUC} = \frac{1}{n_1 n_0} \sum_{r=1}^{n_1} \sum_{q=1}^{n_0} \left[ \mathbf{1}[a_r > b_q] + \frac{1}{2} \mathbf{1}[a_r = b_q] \right]. \quad (28)$$

*Source AUC* uses the source scores: it is the probability that a random BH source scores

above a random NS source, and does not depend on the threshold. *Observation AUC* uses all observations and is used within states.

- **Proper scores (v6g):** class-balanced source-level log loss and Brier score, with source scores clipped to  $[0.01, 0.99]$ . Unlike AUC, they cannot saturate at a perfect ranking.

### 8.3 Class-stratified source bootstrap

The source is the unit of generalisation, so uncertainty is estimated by resampling sources while keeping the OOF scores fixed. The draws are identical in every round:

1. the BH and NS source names are sorted, and `rng = numpy.random.default_rng(42)`;
2. draw  $b$  ( $b = 1, \dots, 2000$ ) takes  $S_1$  BH sources and  $S_0$  NS sources with replacement,

$$\mathcal{D}^{(b)} = (\text{rng.choice}(\mathcal{S}_{\text{BH}}, S_1), \text{rng.choice}(\mathcal{S}_{\text{NS}}, S_0));$$

3. every metric is recomputed on the drawn multiset, where a source drawn twice counts twice and brings all its observations;
4. the 95% interval is the percentile interval  $[Q_{0.025}, Q_{0.975}]$  of the 2000 values.

**Paired differences.** Two models evaluated on the same sources use the same draws, so the difference is computed draw by draw,  $\Delta^{(b)} = M_A^{(b)} - M_B^{(b)}$ . The script also stores `frac_first_better`, the fraction of draws with  $\Delta^{(b)} > 0$ . **Decision rule** (planned from v4 on): a difference is *detected* only if its 95% interval excludes 0. An interval that touches 0 is not detected. Each metric is judged separately; source BA and source AUC are the headline metrics. Code: `09_bootstrap_sources.py`, `v3lib.bootstrap`, `v4lib.paired_bootstrap`, `v7lib.boot_draws`.

**What the interval contains.** Models are not refitted inside the bootstrap, so the interval reflects which sources happen to be in the sample, not the variability of fitting or selection. With 7 BH sources, one BH source changes the source BA by  $1/14 \approx 0.07$ , so many interval endpoints sit exactly at 0.

**State-stratified variant (v8lib.strat\_boot).** For an observation-level metric within a state, sources are drawn from the whole sample (the same draws as above), and the metric is computed on the drawn sources' observations in that state. A draw without BH or without NS observations in the state gives NaN and is ignored; the number of valid draws is reported. Differences between states and between models are paired by draw.

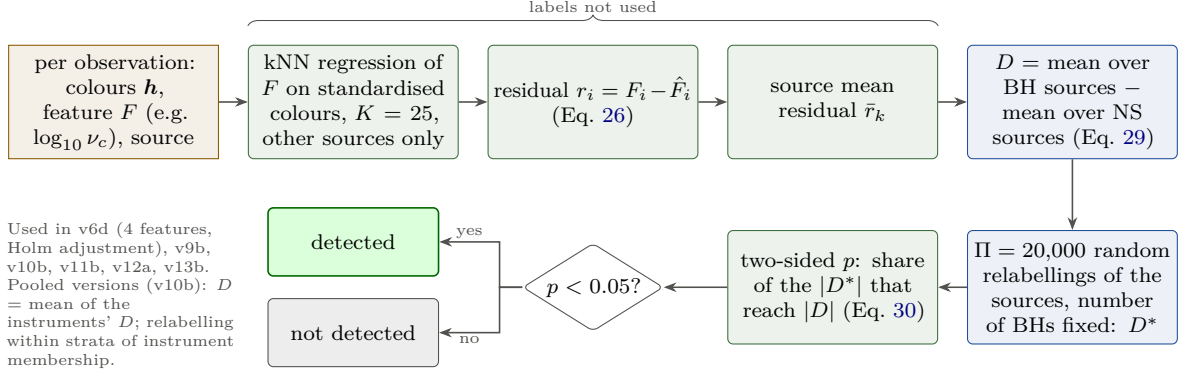
**Pooled variant (v10a, v10lib.pooled\_boot).** RXTE and NICER sources within  $0.1^\circ$  of each other are mapped to one canonical name (`v10lib.canonical_map`). Draws are taken over the union of sources. In each draw each component (RXTE development, RXTE external, NICER) computes its own state-stratified gain, and the pooled statistic is their equal-weight mean. A source observed by two instruments is therefore drawn, or not, in both at once.

### 8.4 Colour-conditional source permutation test

The question is: at matched colour, does a timing feature  $F$  differ between BH and NS sources? Figure 14 shows the procedure.

With the source mean residuals  $\bar{r}_k$  of Equation (26), the statistic is

$$D = \frac{1}{S_1} \sum_{k \in \mathcal{S}_{\text{BH}}} \bar{r}_k - \frac{1}{S_0} \sum_{k \in \mathcal{S}_{\text{NS}}} \bar{r}_k. \quad (29)$$



**Figure 14:** Colour-conditional source permutation test. The colour dependence of the feature is removed without labels, so the class enters only through  $D$  and its permutation distribution; sources, not observations, are relabelled.

If, given colour,  $F$  does not depend on the class, the source labels are exchangeable. The labels are therefore reassigned at random  $\Pi = 20\,000$  times, keeping the number of BH sources fixed (`v6lib.perm_test`, seed 42), and

$$p = \frac{1 + \#\{\pi : |D^{*(\pi)}| \geq |D| - 10^{-15}\}}{1 + \Pi}. \quad (30)$$

Because sources are permuted as a whole, the correlation between observations of one source is respected. When several features are tested together (v6d:  $T_1^*$ ,  $T_2^*$ ,  $T_3^*$ ,  $\log_{10} \nu_c$ ), Holm-adjusted  $p$ -values are also reported (`v6lib.holm`). On synthetic data with no effect, 3.3% of 60 tests gave  $p < 0.05$ .

**Pooled across instruments (v10b, v12a).** Each instrument has its own residuals and  $D_i$ ; the statistic is their mean, and each source keeps one label across instruments. The planned null permuted labels over the union of sources. A synthetic test, run before any v10b statistic, showed that this widens the null, so labels are permuted *within strata of instrument membership* (RXTE only, NICER only, both), keeping the number of BHs per stratum fixed (`v10lib.pooled_perm`). On synthetic data this gave 6.5% of  $p < 0.05$  under no effect (200 sets) and power 0.98 for a 1.5 SD shift.

## 8.5 Conformal sets and observation budgets (v6e, v6f)

**Mondrian conformal sets (`v6lib.conformal_sets`).** For a new source with score  $s$ , class-wise  $p$ -values are computed against calibration sources of each class:

$$p_{\text{BH}}(s) = \frac{1 + \#\{k \in \text{BH} : \bar{s}_k \leq s\}}{1 + n_{\text{BH}}}, \quad p_{\text{NS}}(s) = \frac{1 + \#\{k \in \text{NS} : \bar{s}_k \geq s\}}{1 + n_{\text{NS}}},$$

and the prediction set is  $\{y : p_y(s) > \varepsilon\}$  for  $\varepsilon \in \{0.1, 0.2\}$ . The set can be {BH}, {NS}, both (“cannot tell”) or empty. Two versions are used: a LOSO version on S1 (the other sources’ OOF scores calibrate) and a split version for the external set E (the S1 OOF scores calibrate the frozen-model scores).

**Observation budget (`v6lib.budget_curves`).** For  $k \in \{1, 2, 3, 5, 8, 13, 21\}$  and 500 draws (seed 42), each source’s score is the mean OOF score of  $\min(k, n_k)$  of its observations, drawn without replacement, using the S3 scores (up to 1329 observations per source). The curves show source BA and AUC against  $k$ .

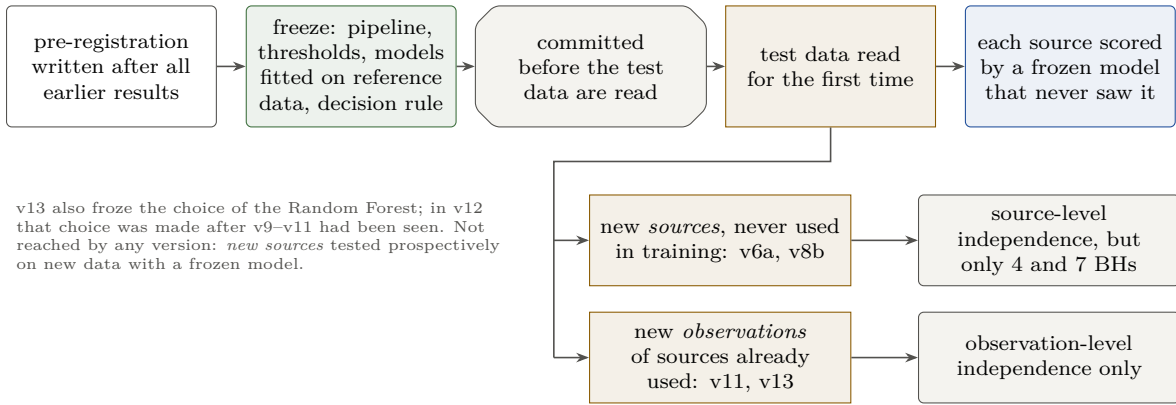
**Agreement and reliability.** The agreement between two models is the Spearman correlation of their OOF scores and the fraction of identical decisions. The bridge check of v4b2 allowed  $H_R$  to replace  $H$  only if the paired intervals contained 0 and the scores correlated at Spearman  $\geq 0.90$ . For split-half reliability, a feature is computed from odd and even rows or segments separately, and the correlation is stepped up with Spearman–Brown,  $2\rho/(1 + \rho)$ .

## 8.6 Frozen and prospective tests

For the external and new-observation tests, everything that could adapt to the test data was fixed and committed before the data were read (Figure 15 and Table 18). Two kinds of independence are distinguished:

- *source-level*: the test sources were never used in training (v6a, v8b);
- *observation-level*: new observations of sources already used (v11, v13).

For NICER BHs only the second was possible, because every dynamically confirmed BH with usable timing data had already been used.



**Figure 15:** Frozen and prospective protocol.

**Table 18:** What was frozen, and when (commit of the plan).

| Test | Frozen before the test data were read                                                                                                    | How the models are applied                                              | Record                        |
|------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------|
| v6a  | $H_R$ -LR, $H_R$ +T-LR and CCTLR( $H_R$ ) fitted once on S1; imputation medians from S1                                                  | every external source scored by the same S1 models                      | plan v6                       |
| v8b  | the v6 models; the v7a state rule                                                                                                        | external set X                                                          | plan v8 (a481fa0)             |
| v11  | NICER feature pipeline, state thresholds, models fitted on the v10 data, $\nu^* = 1.3007$ Hz                                             | out-of-source for the v9 sources; all-data model for never-used sources | plan 6fea016; $\nu^*$ 58ded89 |
| v13  | v12 pipeline with 3C50 correction and screening, RF as the primary model, models fitted on the 534 v12 observations, $\nu^* = 1.3361$ Hz | out-of-source                                                           | plan 42fe1d2                  |

## 8.7 Implementation checks and how verdicts are labelled

From v4 on, every round defines *implementation checks* (IC) that must pass before any result is looked at. Examples:

- earlier OOF predictions are reproduced to  $10^{-8}$ – $10^{-15}$ ;
- synthetic signals are recovered by the timing estimators within 10%;

- pure-Poisson data give  $|z| < 3$ ;
- known sources behave as expected (XTE J1118+480 hard-like, Z sources soft-like);
- results agree with independent tools (XSPEC, `astropy`).

The unit tests are `scripts/test_v4lib.py`–`test_v11lib.py`, and their logs are in `logs/`. A failed check stops the sub-analysis (the scripts exit with code 3) or downgrades it to exploratory.

Each sub-analysis declares one *primary* comparison in its plan; the others are *secondary* or *descriptive* and are not used to claim a difference. Comparisons added after results were seen are labelled *post hoc*. No family-wise correction is applied across descriptive comparisons; where there are many (v4a: 76 combinations  $\times$  3 metrics), the reports state how many intervals would exclude 0 by chance and draw no conclusion from them. Within the v6d permutation family, Holm adjustment is reported.

## 9 Running, logging and reproducing the workflow

### 9.1 Software environment

- **Python.** Python 3.12.10 in a virtual environment on Windows (`logs/environment.txt`); the re-checks used macOS with Python 3.12.14. Packages are pinned in `requirements-lock.txt`, among them numpy 2.4.4, pandas 2.3.3, scipy 1.17.1, scikit-learn 1.8.0, astropy 7.2.0, matplotlib 3.10.9, joblib 1.6.0 and requests 2.33.1.
- **HEASoft.** HEASoft 6.37.1 runs in WSL Ubuntu 22.04, in a conda environment `henv` created by `scripts/install_heasoft_wsl.sh` (Miniforge, then `mamba create -n henv heasoft` from the HEASARC conda channel). It is called from Python with `wsl -d Ubuntu-22.04 -u root -- bash <script>` in three places only: the XSPEC cross-check (v2), `nh` and XSPEC `tbabs` (v8d), and `nibackgen3C50` with the NICER CALDB (v12, v13). On Linux the two shell scripts can be run directly.
- **Other tools.** `curl` for range requests (NICER, RXTE event files) and for downloads inside WSL.

### 9.2 Environment variables

| Variable                         | Effect                                                                                                        |
|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>XRB_VERSION</code>         | selects the <code>config.py</code> block and the output folders (v1 if unset); most scripts set it themselves |
| <code>XRB_PY</code>              | Python interpreter used by the run scripts (default <code>~/venvs/xrb-pilot/bin/python</code> )               |
| <code>XRB_EXTERNAL_RAW</code>    | folder for large raw data outside the repository (default <code>~/xrb-pilot-data/raw</code> )                 |
| <code>V4_JOBS</code>             | parallel jobs of the v4a search (default 16)                                                                  |
| <code>V7C_SMOKE, V9_SMOKE</code> | quick test runs on a few sources or groups (outputs written to <code>*_smoke.csv</code> )                     |
| <code>V9_THREADS</code>          | parallel (source, group) pairs in the v9 NICER fetch (default 8)                                              |
| <code>V8_SKIP_HF</code>          | skip the event-mode part of <code>44_v8_analysis.py</code>                                                    |

### 9.3 Run scripts and order

Each round has a Bash and a PowerShell driver in the repository root. They run the unit tests first, then the scripts in order, writing one log per script to `logs/<v>/`. Table 19 gives the order. Rounds build on each other, so they must be run in sequence; downloads are cached, so a re-run fetches only what is missing.

**Table 19:** Run scripts (.sh; the .ps1 versions are equivalent) and the scripts they call, in order.

| Driver                | Scripts called                                                                     | Notes                           |
|-----------------------|------------------------------------------------------------------------------------|---------------------------------|
| run_all.sh            | 03, 04, 05 (v1 only), 06, 07, 08, 09; 10 with --xspec                              | --v2 switches to v2             |
| run_v3.sh a b c       | a: 11; b: 12; c: 13, then 03/04/06 with XRB_VERSION=v3c, then 14                   | verify the candidate list first |
| run_v4.sh             | test_v4lib; a: 16; burst: 17, 18; b3: 19; b2: 20, 21, 22; b1:                      | b1 downloads about 2 GB         |
| test a burst b3 b2 b1 | 23, 24, 25                                                                         |                                 |
| run_v5.sh             | test_v5lib, 26, 27                                                                 | 26 stops if a check fails       |
| run_v6.sh             | test_v6lib, 28, 21 (XRB_VERSION=v6), 29, 30                                        |                                 |
| run_v7.sh             | test_v7lib, 31, 32, 33, 34, 35, 04 (v7b2), 31<br>--sample v7b2, 36, 37, 38, 39, 40 | 39 takes about 2 h              |
| run_v8.sh             | test_v8lib, 41, 21 (v8b), 42, 43, 44, 45, 46                                       | 45 needs HEASoft (WSL)          |
| run_v9.sh             | test_v9lib, 47, 48, 49, 50, 51                                                     | prefixes about 14 GB            |
| run_v10.sh            | test_v10lib, 52, 53, 54                                                            | tails about 24 GB               |
| run_v11.sh            | test_v11lib, 55, 56, 57                                                            | streams about 40 GB, not stored |
| run_v12.sh            | 58, 59                                                                             | 3C50 through WSL                |
| run_v13.sh            | 60, 61, 62, 63                                                                     |                                 |

**Stops.** A script exits with code 3 when a planned check fails: implementation checks (26, 43, 44, 45), the Z-source and XTE J1118+480 sanity gates (34, 49, 59, 63), the reproduction tolerance (37) and the overlap checks of the NICER selections (55, 60). The drivers stop at that point, except after 43: the recorded v8c run failed its check, so v8c is exploratory and the driver continues.

## 9.4 Logs and checkpoints

- `logs/<v>/<script>.log`: the standard output of each script, with counts, statuses and timings.
- `logs/progress.jsonl` and `reports/progress.json`: one entry per completed stage (`common.progress`).
- `logs/downloads.jsonl`: the download log of Section 2.5. It has 50 777 lines: 50 773 JSON records, three fragments of interrupted concurrent writes and one empty line.
- `logs/04_errors.log`: tracebacks of RXTE candidates that could not be fetched or read.
- `data/v11/observations_partial.jsonl`, `data/v13/observations_partial.jsonl`, `data/v12/bkg3c50_partial.jsonl`, `data/v13/bkg3c50_partial.jsonl`: checkpoints of the long NICER steps; an interrupted run resumes from them.
- `logs/v11/nu_star_frozen.txt`, `logs/v13/nu_star_frozen.txt`: the frozen  $\nu^*$  values.

## 9.5 Plans, decision log and branches

- **Plans.** `reports/preregistration.md` (v1) and `reports/preregistration_v2.md` to `_v13.md`. Each is a time-stamped, internally pre-specified analysis plan, written after all earlier results and before the round’s data were downloaded. It fixes the samples, features, models, primary comparison and decision rule. The matching block of `config.py` is committed with it.
- **Decision log.** `reports/decision_log.md` records every deviation, bug fix and added comparison, with the time and whether it was made after the round’s results were seen.
- **Branches.** From v3 on, each round was developed on a branch `vN-analysis` and merged into `master` through a pull request (v3 to v13, then a v14 wrap-up). The commit SHAs of the frozen plans are cited in Table 18.

- **Reports.** Each round has a report `reports/report_vN_zh-TW.md`, ending with a “re-run” section that lists the exact order, durations and download volumes.

## 9.6 Data locations

- **In git:** source tables; candidate tables; every tried observation with its status and reason (`data/*/observations.csv`); processed features (`features.npz`, `*_features.csv`); out-of-fold predictions; result tables (`results/`); figures; logs.
- **Not in git** (`.gitignore`): raw RXTE products of v1–v3 and the v8 event files in `data/raw/`; under `XRB_EXTERNAL_RAW`, the v4b1 files, the NICER prefixes (v9), tails (v10) and the v13 cleaned files of faint observations; the 3C50 spectra in `~/xrb-pilot-data/nicer_bkg_spectra/`; the NICER CALDB in `~/xrb-pilot-data/caldb/`. Everything can be fetched again with the scripts.

## 9.7 How to reproduce a result

1. Clone the repository, create a Python 3.12 virtual environment and run `pip install -r requirements-lock.txt`.
2. For v2 (XSPEC), v8d, v12 and v13, install HEASoft in WSL with `scripts/install_heasoft_wsl.sh`, and for v12–v13 unpack the NICER CALDB into `~/xrb-pilot-data/caldb`.
3. Run the drivers in order (Table 19). Each script re-downloads only missing files, re-applies the frozen rules and rewrites its outputs.
4. Compare: re-downloaded files can be checked against the SHA256 values in `logs/downloads.jsonl`, and the regenerated tables against the committed ones (`git diff`).

This was done once in full for v1–v2 on a second machine (macOS). All 2689 raw files matched their SHA256 values. `features.npz` agreed to  $\leq 10^{-16}$ , LR scores were identical, single RF scores differed by at most 0.006 (source means by at most 0.001), and all predicted labels and accuracies were identical.

## 9.8 Script index

**Table 20:** Analysis scripts (`scripts/`), what they do and their main outputs.

| Script                    | Purpose                                                                | Main outputs                                                                                                 |
|---------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 01, 02                    | inspect the archive; download MissionLongData catalogues               | <code>data/cache/</code> , <code>data/raw/catalogs/</code> (not in git)                                      |
| 03_select_observations.py | source table and time-stratified candidates                            | <code>sources.csv</code> , <code>observation_candidates.csv</code> , <code>attrition_catalog.csv</code>      |
| 04_fetch_process.py       | download Standard Products, dead time, quality rules, rebinning        | <code>features.npz</code> , <code>observations.csv</code> , <code>attrition_download.csv</code>              |
| 05, 06                    | two-source data-flow check; dataset checks                             | <code>reports/step2_fits_check.md</code> , <code>results/data_checks.md</code>                               |
| 07, 08, 09                | LOSO (Dummy, LR, RF) and leaky split; error analysis; source bootstrap | <code>oof_predictions_loso.csv</code> , <code>fold_loso.csv</code> , <code>bootstrap_source_level.csv</code> |
| 10_xspec_crosscheck.py    | XSPEC check of the background subtraction (WSL)                        | <code>results/v2/xspec_crosscheck.csv</code>                                                                 |
| 11, 12                    | v3a increments; v3b 3–25 keV                                           | <code>results/v3/</code>                                                                                     |
| 13, 14                    | v3c candidate catalogues and scores                                    | <code>data/v3c/</code> , <code>results/v3c/</code>                                                           |
| 15                        | colour-colour summary figure                                           | <code>figures/final/</code>                                                                                  |
| 16_v4a_algorithms.py      | eight-algorithm search, nested selection                               | <code>results/v4/v4a_*</code>                                                                                |
| 17, 18                    | burst detection; burst exclusion                                       | <code>results/v4/burst_*</code>                                                                              |
| 19_v4b3_hexte.py          | HEXTE colours                                                          | <code>data/v4b3/processed/hexte_features.csv</code>                                                          |
| 20–22                     | v4b2: position-matched sources, download, analysis                     | <code>data/v4b2/</code> , <code>results/v4b2/</code>                                                         |

| Script                                                                                               | Purpose                                                                                                                                           | Main outputs                                                                                                                              |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 23–25<br>26, 27                                                                                      | v4b1: all epoch-5 pointings<br>Standard-1 timing features; v5 analysis                                                                            | data/v4b1/, results/v4b1/<br>data/v5/processed/timing_features.csv,<br>results/v5/                                                        |
| 28–30                                                                                                | external sources; all-pairs timing and $\nu_c$ ;<br>v6a–h                                                                                         | data/v6/, results/v6/                                                                                                                     |
| 31–33                                                                                                | MINBAR matching, replacement,<br>exclusion                                                                                                        | results/v7b1/, data/v7b1/                                                                                                                 |
| 34_v7a_states.py<br>35, 36                                                                           | variability states<br>persistent BHs                                                                                                              | results/v7a/states.csv<br>data/v7b2/, results/v7b2/                                                                                       |
| 37_v7c1_repro.py<br>38, 39<br>41, 42                                                                 | reproduction of de Beurs et al.<br>MAXI selection, cuts and models<br>new BHs; states and timing of the<br>external set                           | results/v7c/v7c1_*<br>data/v7c/, results/v7c/v7c2_*<br>data/v8b/                                                                          |
| 43_v8c_events.py<br>44, 45<br>47–49                                                                  | event-mode features<br>v8a–c analysis; $N_H$ correction (WSL)<br>NICER selection, prefix features,<br>analysis                                    | data/v8c/processed/hf_features.csv<br>results/v8*/, data/v8d/<br>data/v9/, results/v9a/, results/v9b/                                     |
| 50<br>52–54                                                                                          | MAXI colour strata<br>NICER full files; RXTE $\nu_c$ table;<br>pooled tests                                                                       | results/v9c/<br>data/v10/, results/v10a/–v10c/                                                                                            |
| 55–57<br>58, 59<br>60–63<br>40, 46, 51, 64                                                           | new NICER observations; frozen tests<br>3C50 (WSL); corrected analysis<br>third NICER set; frozen v12 pipeline<br>summary figures                 | data/v11/, results/v11/<br>data/v12/, results/v12/<br>data/v13/, results/v13/<br>figures/v7/, figures/v8/, figures/v9/,<br>figures/final/ |
| common.py, spectra.py,<br>config.py,<br>plotstyle.py<br>v3lib.py–v11lib.py                           | download and progress log; Standard-2<br>reader; settings; plot style<br>round libraries (representations, models, –<br>bootstrap, timing, NICER) | –                                                                                                                                         |
| test_v4lib.py–<br>test_v11lib.py<br>v12_bkg3c50.sh,<br>v8d_heasoft.sh,<br>install_heasoft_wsl.<br>sh | implementation checks on synthetic<br>data<br>HEASoft steps in WSL                                                                                | logs/v*/test_*.log<br>–                                                                                                                   |

## A File schemas

The columns below were read from the committed files at commit 666c2e0. “Rows” is the row count of the example file named. Files of other rounds with the same name have the same columns, unless noted.

**Table 21:** Schemas of the main tables and arrays.

| File (example; rows)                      | Columns and meaning                                                                                                                                                                                                                                                                |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>RXTE</i>                               |                                                                                                                                                                                                                                                                                    |
| data/v2/sources.csv (33)                  | source_id, name, aliases, ra_deg, dec_deg, label, evidence, reference_url, status, coord_origin, n_eligible, included. label is BH or NS; evidence and reference_url give the labelling evidence; included is False if fewer than the minimum number of eligible pointings remain. |
| results/v2/attrition_catalog.csv (33)     | source_id, catalog_rows, after_dedup, after_epoch5_window, after_exposure, after_std1rate, after_pointing: rows left after each eligibility step.                                                                                                                                  |
| data/v2/observation_candidates.csv (7975) | obs_id, source_id, label, time_bin, rank_in_bin, mjd, cat_exposure, cat_npcu, cat_std1rate, pointing_sep_deg: every eligible pointing with its time group and rank.                                                                                                                |

| File (example; rows)                         | Columns and meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data/v2/observations.csv (462)               | <i>Candidate:</i> obs_id, source_id, label, time_bin, rank_in_bin, mjd, cat_exposure, cat_npcu, cat_std1rate. <i>Files:</i> url_base, path_src, path_bkg, path_rsp. <i>Header:</i> date_obs, obs_mjd_start, exposure_s, bkg_exposure_s, pcus, npcu, n_gti, object_hdr, hdr_sep_deg, src_unit, src_class, bkg_class, poisserr, sys_err, backscal_ratio, rsp_detnam, gain_shift_channels. <i>Dead time:</i> dtf, dcor, dtf_xe, dtf_std1, dtf_std1_vle150us, std1_npcu_on, std1_covered_s, std1_vle_rate_per_pcu, deadtime_method. <i>Summary:</i> net_rate_5_25, net_rate_err, net_snr, src_rate_5_25, bkg_rate_5_25, bkg_fraction, hardness_10_25_over_5_10, n_nonpositive_bins. <i>Decision:</i> status (accepted or excluded), reason, in_dataset. |
| results/v2/attrition_download.csv (31)       | source_id, candidates_tried, accepted, bins_empty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| data/v2/processed/features.npz (456)         | rate, err, srate, brate, pl2: $456 \times 45$ float64 in $\text{cs}^{-1} \text{keV}^{-1} \text{PCU}^{-1}$ (net rate, its error, total rate, background rate, folded $\Gamma = 2$ power law); edges: 46 grid edges in keV; obs_id, source_id: strings; y: int64 (1 = BH); F: net 5–25 keV rate per PCU. Rows are in the order of the accepted observations.                                                                                                                                                                                                                                                                                                                                                                                          |
| data/v5/processed/timing_features.csv (8170) | obs_id, source_id, label, bkg_rate_per_pcu, n_rows_in_gti, n_std1_files, n_rows_valid, f_T1..f_T3 (fractional variances), fauto_T1..fauto_T3 (auto-power check), n_on, src_rate_per_pcu, tot_rate_per_pcu, T1, T2, T3, T_total (signed rms), T1_auto..T3_auto, status, in_v2, in_v4b2, in_v4b1.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| data/v6/processed/timing_v6.csv (8350)       | obs_id, source_id, label, n_rows_in_gti, n_rows_valid, n_rows_pairs, n_rows_single, T1s, T2s, T3s, Ts_total, nu_c, lb0..lb8 (log-bin variances), odd/even halves T1s_odd..T3_v5_even, status, in_v2, in_v4b2, in_v4b1, in_v6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| results/v7a/states.csv (456)                 | obs_id, source_id, label, n_rows_in_gti, n_rows_burst_removed, n_rows_used, rms, state, c1, c2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| data/v4b3/processed/hexte_features.csv (456) | obs_id, source_id, label, mjd, status, reason, respfile, ancrfile, deadapp, exposure_src, exposure_bkg, X_25_40, X_25_40_err, X_40_60, X_40_60_err, nchan, F_PCA_16_25, F_PCA_16_25_err, X1, X2, X1_den_snr, X2_den_snr.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| data/v8c/processed/hf_features.csv (311)     | obs_id, source_id, label, sample, role, bkg_rate_per_pcu, n_event_files, n_events, n_seg; for each band LC, HF1, HF2, HF3, NULL: <band>_var, <band>_se, <band> and the mean-of-ratios variants; NULL_z, HF1_z..HF3_z, mean_n_on, src_rate_per_pcu, tot_rate_per_pcu, exposure_s, status, n_burst_intervals.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <i>NICER</i>                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| data/v9/sources.csv (140)                    | name, label, ra, dec, evidence, pos_origin, excluded_confused, eligible, included.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| data/v9/observation_candidates.csv (3848)    | obsid, source, label, time_bin, rank_in_bin, mjd, exposure, num_fpm, ym, url (ym is the archive month folder). v11 and v13 add group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| data/v11/observations.csv (588)              | <i>Candidate:</i> obsid, source, label, group, time_bin, rank_in_bin, mjd, exposure_cat. <i>Reading:</i> n_events, bytes, file_bytes, read_complete, sha256, url. <i>Segments:</i> n_bursts, n_seg, n_seg_burst_removed, good_time_s, rate_2_10, mean_n_mpu. <i>Colours:</i> cA, cB, cC, c1, c2. <i>Timing:</i> T1_var, T1_se, T1 and the same for T2, T3, STATE, WHITE; rms_nuc_band, nu_c, lb0..lb12, lb0_se..lb12_se. <i>Decision:</i> state, status (ok, missing or excluded with reason), bkg_ratio_12_15. The v9 file has url, bytes, file_bytes, stop, rows_read instead of the streaming fields, and no group, bkg_ratio_12_15 or lb*_se.                                                                                                   |

| File (example; rows)                                          | Columns and meaning                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data/v12/processed/<br>bkg3c50.csv (503)                      | obsid, source, label, sample, rate_2_10, rc (exit code), T_A, T_B, T_C, T_T (total band rates), B_A, B_B, B_C, B_T (background band rates), f_A, f_B, f_C, f_T (fractions), exposure_3c50, status.                                                                                                                                                                                    |
| data/v12/processed/<br>nicer_corrected_<br>screened.csv (534) | obsid, source, label, c1, c2, cA, cB, cC, T1, T2, T3, STATE, nu_c, rate_2_10, state (after correction), sample (v10 or v11), bkg_status, T_T, f_A, f_B, f_C, f_T, f_used.                                                                                                                                                                                                             |
| MAXI                                                          |                                                                                                                                                                                                                                                                                                                                                                                       |
| data/v7c/c2_sources.csv<br>(134)                              | name, cls (BH, NPNS, PSR), maxi_id, maxi_name, n_points_raw, n_points_cut, included, evidence, match_sep_deg.                                                                                                                                                                                                                                                                         |
| data/v7c/maxi_points.<br>csv (132057)                         | mjd, f, e (total), L, eL, M, eM, H, eH (2–4, 4–10, 10–20 keV and errors), source (MAXI J-identifier), name, cls.                                                                                                                                                                                                                                                                      |
| data/v8d/nh_hi4pi.csv<br>(100);                               | name, ra, dec, avnh, avwnh, map; nh_1e22, L, M, H (photon flux per band for each $N_H$ of the grid).                                                                                                                                                                                                                                                                                  |
| tbabs_transmission.csv<br>(201)                               |                                                                                                                                                                                                                                                                                                                                                                                       |
| Model outputs                                                 |                                                                                                                                                                                                                                                                                                                                                                                       |
| results/v2/oof_<br>predictions_loso.csv<br>(5472)             | representation, model, fold, source_id, obs_id, true_label, BH_score, predicted_label: one row per observation and model.                                                                                                                                                                                                                                                             |
| results/v13/v13_frozen_<br>predictions.csv (792)              | as the OOF table plus<br>raw_score, threshold, name, state, nu_c, f_used.                                                                                                                                                                                                                                                                                                             |
| results/v13/v13a_gain.<br>csv (21)                            | analysis, comparison, point, ci2_5, ci97_5, n_valid_draws, role (PRIMARY or descriptive),<br>verdict, name, metric, frac_first_better. Other bootstrap tables use the same pattern.                                                                                                                                                                                                   |
| Logs                                                          |                                                                                                                                                                                                                                                                                                                                                                                       |
| logs/downloads.jsonl<br>(50773 records)                       | Five record types: url, path, bytes, sha256, utc (47073, common.download); the same plus tool (1707: curl downloads and 3C50 inputs, whose path says they were deleted); prefix records with file_bytes, rows, naxis2, stop (799, v9); tail records with offset, file_bytes, complete (192, v10); streamed records without path but with file_bytes, rows, complete (1002, v11, v13). |
| logs/progress.jsonl                                           | utc, stage, details: one line per completed stage.                                                                                                                                                                                                                                                                                                                                    |

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